

A Survey on the Unmanned Aircraft System Traffic Management

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The Unmanned Aircraft System Traffic Management (UTM) system is a set of services offering an automated management of the airspace and thus providing safe and secure Unmanned Aerial Vehicle (UAV) flights in both controlled and uncontrolled airspace. Controlled airspace refers to the portion of the airspace that is under the authority of Air Traffic Control (ATC) and where separation services are offered, while uncontrolled airspace refers to the portion of airspace where aircraft are not regulated by ATC. This article is a comprehensive survey of the existing UTM's development efforts with a focus on the different UTM's architectures, the provided services, the used communication technologies and the decision-making process within UTM's. We firstly review the different UTM architecture and propose a novel UTM taxonomy based on high-level qualitative criteria. Secondly, we detail the services provided by UTM's with an emphasis on the used technologies in the identification, the surveillance, the monitoring, and the deconfliction services. Effective decision-making is crucial, particularly in emergency scenarios such as Air-to-Ground (A2G) communication loss, battery or motor malfunction, or encountering aerial obstacles, among other potential hazards. Despite its significance, the UTM decision-making process is not enough considered in the literature and especially in UTM surveys. We analyze and compare in this article both the centralized and decentralized UTM decision-making. Centralized decision-making is not conducted in real-time and primarily relies on Air-to-Ground (A2G) communication. In the decentralized case, the decision-making process primarily relies on communication and collaboration among UAVs with varying degrees of autonomy. We show in this paper that centralized decision-making may encounter issues with packet loss and imperfect data, which can negatively impact the quality of decision-making. We also highlight that the decentralized decision-making may also face challenges related to security and scalability, which can hinder its effectiveness. Finally, evaluating the performance of UTM's on a real environment raises several challenges and the simulation is a cost-effective alternative. Hence, we provide a summary of the existing UTM's simulators and discuss their main features.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Computer systems organization** → **Distributed architectures**; • **Networks** → *Network types*; • **Applied computing** → **Aerospace**;

Additional Key Words and Phrases: UTM, UAV, architecture, communication, performance, services, simulators, decision-making

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1 INTRODUCTION

The small size, low energy consumption and maintenance cost of the UAVs have promoted their use for recreational purposes and civil applications. The **Federal Aviation Administration (FAA)**, which is responsible for the regulation and oversight of civil aviation within the United States, forecasts that the recreational small drone market will saturate at around 1.81 million units over the year 2026 and the commercial drone fleet will likely be at around 858,000 in the US [5]. In a similar context, the number of consumer leisure drones will be about 7 million in 2050 and 400,000 for governmental and commercial purposes in Europe [93]. Agriculture and forestry, logistics and transportation, surveillance and monitoring, construction and infrastructure inspection, are the sectors that may be covered by the UAVs, to cite a few [85]. In the European Drones Outlook Study [93], the **Single European Sky ATM Research (SESAR)** reported that in Europe, almost 125,000 drones by 2035 will be needed to cover 500 hectares a day for agricultural activities like harvest and fertilizing. The energy demand of drones for utility lines and pipelines is expected to be almost 1,000 fixed wing drones to inspect approximately 8 million kilometers every 3 to 18 months. Similarly, the demand for E-commerce and delivery activities is predicted to be around 7,000 drones to deliver almost 200 million parcels in 2035 of goods like pharmaceuticals, food, and electronics. To fully unlock the potential of the UAV market, it is urgent to address and overcome various issues and challenges.

The risks of collisions, the substantial magnitude of the data generated by UAVs, and their dynamic nature pose significant difficulties in managing the airspace with the traditional **Airspace Traffic Management** system (ATM). Several incidents caused by drones have been noted namely a drone crash on the White House lawn in 2015, a drone used to record video crashed into the stands in Virginia Motorsports Park, injuring several people in attendance in 2013 and the serious risk of collision between a drone that came close to an Airbus A320 when landing at Heathrow in 2014 [90]. Academic, commercial, and government efforts are made to create adequate UTM systems. About 200 million EUR are invested on the research and development related to the creation of UTMs in Europe [93]. Recently, numerous studies have dealt with UTMs in several aspects. Above all, most of the work has focused on developing UTMs and creating high-level architectures trying to bring together all entities participating in such a complex system [12, 26, 41, 62–64, 75, 87, 88]. The safety aspect, well considered in the development of the UTM architectures, justifies the need of robust communication systems with more targeted technologies and deconfliction methods [3, 60, 72, 97]. However, the UTM is still in a state of childhood, only a few implementations exist [10, 31, 36] and some developed simulators: [29, 42, 102]. To the best of our knowledge, there are only a few surveys that have coped with UTMs: [19, 22, 87, 98]. These surveys can be categorized into those that focus on communication technologies [19], UTM development, regulations and laws [98], and those that address **Urban Air Mobility (UAM)** [22, 87]. [22] primarily delineates the means to regulate the movement of air traffic within urban airspace by leveraging a UAM system. It outlines diverse strategies for configuring the airspace and elucidates the various factors that can potentially impact the airspace design. It is worth noting that the communication aspect between the UAVs and the UAM has not been expounded upon. [19] delves into the communication technologies and connectivity prerequisites for various categories of aerial vehicles. The UTM is only considered as a specific instance of a connectivity application. [87] provides an overview of several concepts related to the UTM, including the conceptual models for designing the structure of an urban airspace, cyber-security risks associated with UAVs, and the cellular communication technologies that can be employed in UTMs. The analysis focuses on a comparison limited to the US UTM and the U-space, without taking into account other UTM systems that have already been developed. [98] primarily focuses on analyzing the issues and future directions of existing policies and management technologies for low-altitude UAV flying. These

Table 1. The UTM Surveys

Survey	Reference	Concept of interest	Contribution(s)	Future challenges
(Bauranov and Rakas, 2021)	[22]	- Urban airspace - UAM	- identification of factors that influence the design of the airspace: physical, noise and operational factors - discussion of the different ways to design the airspace: corridors, cells, lanes and tubes - presentation and comparison of governmental and industrial initiatives to design the UTM and UAM systems	a set of recommendations for UAM research: - research of risk, - research of new tracking and communication technologies - more interest for noise factors
(Baltaci et al., 2021)	[19]	-Communication -UTM	- classification of aerial platforms following the connectivity relation: end point or provider - communication requirements of eVTOLs - overview of the UTM as a use case of connectivity application of eVTOLs and UAVs - cellular network 5G and the UTM - evaluation of communication technologies of UAVs	-artificial intelligence: Intelligent inter-UAV routing, vertical, horizontal handoff prediction vs. real-time decision making -network capacity -energy efficiency -a unified architecture for UTM networks -cross-border flights
(Shrestha et al., 2021)	[87]	-UAM -UTM	- presentation of airspace management: ICAO classification, 4 concept designs (full mix, layered, zones and tube concepts) - comparison of US UTM and U space: ConOps, architecture, implementation levels - overview of communication technologies: 4G/LTE, 5G and 6G - safety in the UTM: collision avoidance - security in the UTM: attacks	- the concept of global UTM: adoption of open source UTM
(Xu et al., 2020)	[98]	-Urban areas -Regulations -Civilian UAVs -UTM	- analysis of restrictions in VLL applied on UAVs to ensure safety: maximum flying height, geofence, segregated airspace - revise regulations to motivate UAV industry development - analysis of regulation policies: identification, real name registration, UTM, USS - air routing planning	-to develop quantifiable geofence constraints -GIS and RS technology

include restrictions imposed on the structure of the airspace, such as maximum flying height and geofences, as well as active regulations for UTM systems, such as the requirement for registration and identification of all UAVs operating within the airspace. In Table 1, we present with more details these reviews. Likewise, our contributions to the literature encompass a range of aspects, which can be listed as follows: (i) Addressing fragmented information: Currently, there is a lack of comprehensive papers that bring together all the articles pertaining to UTMs, including their architectures, decision-making processes, simulators, services, communication links, challenges, advancements, and issues. In our review, we aim to fill this gap by consolidating all relevant information; (ii) Comprehensive comparison of architectures: We present and compare almost all architectures found in the literature, with a particular focus on the centralized and decentralized approaches. This comparison allows us to identify the strengths and weaknesses of each architecture, offering valuable insights; (iii) A thorough comparison of decision-making approaches: We also provide a thorough comparative analysis between centralized and decentralized decision-making in UTMs. This analysis sheds light on the distinct characteristics, advantages, and challenges associated with each approach; (iv) In-depth study of communication technologies: Furthermore, we conduct a comprehensive study of communication technologies used in UTMs since their inception until the present day. This examination helps us understand the evolution, applications, and potential issues surrounding communication systems in UTMs. In Table 2, we provide a brief comparative study of existing surveys with our review.

The rest of this paper is organized as follows: In Section 2, we give an overview of the most widely used definitions and terms relative to the UTM. In Section 3, we review the different UTM architectures presented in the literature. Additionally, we present the main features of every

Table 2. Comparison between Existing Surveys and our Survey

Survey Topic	(Bauranov and Rakas,2021)	(Baltaci et al.,2021)	(Shrestha and Kim,2021)	(Xu et al.,2020)	Our survey
Airspace design	√ (especially for urban airspace)	×	√ (especially for urban airspace)	×	√
UTM architectures	overview of several governmental and commercial UTM	×	only US UTM and U-space	overview of some popular UTM	√
UTM services	×	√	remote ID and monitoring	×	√
UTM building blocks	×	×	×	×	√
Regulations	×	×	√ (an overview)	an exhaustive review	√
Decision making in UTM	×	×	×	×	√
Communication: technologies, protocols and messages	communication, navigation and surveillance technologies	√ (exhaustive review)	communication technologies	×	√
UTM: measures of performance	×	×	×	×	√
UTM simulators	×	only one for UTM	×	×	√

architecture, and we discuss their advantages and limits. Thereafter, we study in Section 4 the most important and common services provided by a UTM. We describe the different airspace structures found in the literature in Section 5. Section 6 details the different UTM communication technologies that can support the UTM communication links requirements. We discuss the centralized and decentralized UTM decision-making in Section 7. Afterward, we present in Section 8 the different measures used to assess the performance of the UTM. In Section 9, we summarize the open issues and research directions worthy of new explorations. Finally, Section 10 provides concluding statements.

2 DEFINITIONS AND TAXONOMIES

The UTM concept is currently under standardization by the FAA, **European Union Aviation Safety Agency (EASA)** and several other institutions. However, there is a lack of consensus on defining its key concepts. In this section, we give an overview on the most widely used definitions and terms relative to the UTM.

2.1 Unmanned Aircraft

Various terminologies are employed to refer to unmanned aircraft. Some of them include: **UAV (Unmanned Aerial Vehicle)**, **UAS (Unmanned Aerial System)**, drone, **RPA (Remotely Piloted Aircraft)**, **RPV (Remotely Piloted Vehicle)**, and **RPAS (Remotely Piloted Air System)**. Referring to [32], there are slight variations in the terminologies utilized by different countries and institutions when referring to unmanned aircraft. In fact, the FAA and EASA employ the term UAS. Whereas, the United States Armed Forces uses the term RPA and the British Army uses the term RPAS to designate both the aircraft and the pilot. According to [32], the UAV or drone is defined as *a pilot-less aircraft, flying machine without an on-board human pilot or passengers*. Whereas, according to [3], a UAS is defined as *an unmanned aerial vehicle (UAV) and its controller which use radios to communicate. The remote controller is moving towards long range control and autonomous operation*.

In this paper, we define a UAV as every unmanned aircraft that can be controlled remotely by a pilot or autonomously by an embedded autopilot. They can be used for both civilian and military use cases. Their characteristics and their embedded technologies differ according to their targeted application. A UAS is composed of the UAV and the ground station that performs communication links with it.

2.2 Unmanned Aircraft System Traffic Management (UTM)

The ATM system is responsible for regulating the air traffic within conventional manned aviation. The fundamental goal of the UTM design is to facilitate the integration of the UAVs into the **National Airspace System (NAS)**, particularly in the **Very Low Level (VLL)** airspace, with utmost emphasis on safety and security. The VLL airspace is set to under of 400 feet **Above Ground Level (AGL)**. To achieve this objective, the UTM provides distinct services to the stakeholders that comply with respective country regulations and separate from those offered by ATM. The physical integration of the UAVs in the NAS entails collaborative efforts between the ATM and UTM, encompassing both regulatory and technological endeavors. It takes place at multiple aspects including the following: (i) Airspace design and organization: to share the airspace between manned and unmanned aircraft. (ii) Physical and communication infrastructures: to create adequate areas for takeoff and landing, to create ground control stations for operators to control the UAVs, and to use existing communication infrastructures like the cellular base stations to assure UTM communication links. (iii) Technical and communication standards, protocols: technical standards including the requirements on the UAVs design, performance and embedded technologies, the communication standards to assure reliable and secure communication links and information exchange between the ATM, UTM and all of the involving entities. (iv) Regulations: The authorities provide additional regulations and restrictions to integrate the UAVs as users of the airspace with the already existing manned aircraft. In this section, we provide definitions for UTM and ATM. The **International Civil Aviation Organization (ICAO)** [49] defines the **Air Traffic Management (ATM)** as *the dynamic, integrated management of air traffic and airspace*.

The UTM concept is still in its infancy, and there is a lack of consensus on its precise definition. The FAA [36] defines UTM as *a set of federated services and an all-encompassing framework for managing multiple UAS operations. These services are distinct yet complementary to those offered by the ATM, and are based primarily on the sharing of information between operators on flight intent and airspace constraints*. U-space is the UTM designed for European community. It is defined in [93], as *a set of new services and specific procedures designed to support safe, efficient and secure access to airspace for large numbers of drones*. To make the UTM definition more precise, the **Global UTM Association (GUTMA)** explains in [40] the difference between the UTM concept and the UTM system and defines the two notions. In fact, the UTM as a concept is defined as *a system of stakeholders and technical systems collaborating in certain interactions, and according to certain regulations to ensure air traffic management*. While on the contrary, the UTM as a system is *a concrete technical implementation comprising software, the necessary infrastructure for running the software, and the drones themselves*.

3 UTM ARCHITECTURES

A UTM architecture is based on relationships and information flows between a set of stakeholders. At least, an elemental UTM architecture includes the following entities: (i) The regulator (e.g., FAA): establishes the policies and regulations to ensure airspace safety; (ii) The UAV operator: is the legal entity or a natural person who directs the UAV using a remote pilot or an automatic on-board pilot system. He is responsible to accomplish an operation safely; (iii) The UAVs: autonomous pilot-less aircraft or remotely piloted, having different characteristics and used for different applications; (iv) **UAS Service Suppliers (USSs)**: they represent the liaison point between the UTM and the operators and allow them to benefit from the UTM services; This entity is known as the **U-Space service providers (USSPs)** in the EU and **UTM Service Provider (UTMSP)** in India; (v) **Supplemental Data Service Providers (SDSPs)**: is every entity able to provide access to supplemental data. SDSPs can be categorized in some UTM architectures into a weather data service

Table 3. Commercial UTM

Manufactures	Product	Contribution	Challenges	Reference
Airbus	Sky way	- test a drone parcel delivery services - create an operation drone management system - Airbus A3 Altiscope future ATM framework simulator	- Air transportation - Integration of UAVs with manned aircraft - high level of automation in flight management	[6]
Airmap	Air Space Information and Services	- a complete platform for drones, manufactures, airports to fly in the VLL airspace		[7]
Amazon PimeAir	UTM system best equipped best served	- the access to the airspace is based on material performance - segregated blocks of the VLL airspace		[15]
Astral Aerial	UTM System for Africa	- virtual highways and routes for drones - interaction between pilots with maps		[54]
Unify	Launch pad, Pro app, sentry and connect	- four products to serve users, professionals, ANSP - a total management of the airspace		[94]

provider in charge of giving weather data, a ground risk observation service provider that can provide information about ground and terrain like buildings and data about population density [31]; (vi) Other stakeholders, including public safety entities and the general public. They access the UTM operations data after being authorized for safety, security, and privacy purposes.

3.1 Classification Criteria of the UTM Architectures

Several government initiatives and commercial UTM are produced to cover the need of a system that safely manages UAVs in the airspace. Our proposition entails the classification of the UTM architectures based on two distinct criteria: the entity responsible for providing common information and the distribution of UTM services intended for operators. The primary criterion revolves around the entity accountable for delivering indispensable information to all stakeholders while upholding an adequate standard of quality, integrity, and accuracy. This entity assumes varying designations across different UTM implementations, such as the **Common Information Service (CIS)** within the European U-Space [31] and the **Flight Standards Information Management System (FIMS)** [36] within the US UTM. If this function is performed by a single entity, the system is considered to be centralized. However, in its absence, enabling stakeholders to freely communicate and exchange common data, the UTM is deemed decentralized. The primary UTM systems have centralized architectures, especially for the government initiatives: the US UTM [36], the European U-space [31], the Indian UTM [68] and the UTM of Taiwan [64]. Several companies are racing to create UTM systems. We summarize the characteristics of the most renowned ones in Table 3. Some other works [11, 12] have been oriented towards a more decentralized approach based mainly on blockchain technology used to ensure more security on the information sharing between the different participants. In a distinct facet, [86] attempts to leverage the advantages of **artificial intelligence (AI)** to foster the concept of autonomy and have developed an advanced and more intelligent UTM architecture.

The second criterion pertains to the distribution of the UTM services. It centers on the number of USSs providing UTM services within a particular region. If a single USS offers these services, the UTM is classified as monolithic. Conversely, if multiple USSs provide such services, the UTM is characterized as federated. This section entails a comprehensive analysis of the architectures proposed in existing literature, alongside our proposal of a new UTM taxonomy grounded in the presented high-level qualitative criteria.

3.2 Centralized Architectures

A centralized UTM architecture is based on a central entity having the ability to support relationships and information flows with all the stakeholders. The foundation of a centralized UTM's architecture is argued by the safety issue. A single trustworthy source of critical information is mostly considered more accurate and safer. From the point of view of cost, a single

entity ensuring information sharing is more cost-efficient than multiple entities receiving and distributing information [33]. Nevertheless, centralized systems often encounter performance challenges owing to the possibility of a single point of failure. Various factors, including weather conditions, technical malfunctions, pilot errors, and unexpected events, can lead to abrupt and rapid changes in the behavior and motion of UAVs. These unpredictable dynamics make it challenging for a single entity within the UTM system to accurately predict and anticipate their movements. As the number of UAVs increases, along with their performance disparities and varying technological capabilities, the complexity of the airspace intensifies. Managing such a dynamic and diverse airspace becomes a difficult task, requiring significant resources. Centralized systems heavily rely on precise knowledge of the current state of the airspace to facilitate efficient management and collision avoidance. Accurate and up-to-date information about the position, velocity, and orientation of aircraft is crucial. However, the centralized nature of such systems introduces latency as all communication must pass through a central entity. This latency, combined with sudden changes in velocity or trajectory, can negatively impact the efficiency of the decision-making process. Therefore, the combination of the dynamic behavior of UAVs, the increasing complexity of the airspace with a growing number of UAVs, and the latency introduced by centralized communication poses significant challenges to the effectiveness and efficiency of a centralized UTM system. In this paper, we propose to classify centralized architectures into based on regulator rules, hierarchical, service-oriented and, based on a specific cellular network. We discuss each category based on architectures presented in the literature.

3.2.1 Centralized Architecture based on Regulator Rules. A centralized architecture based on regulator rules relies on a central entity established and managed by a state-owned regulatory body (e.g., FAA), without allowing private organizations to undertake such responsibility. Its objective is to ensure security, reliability, and source integrity for the information. However, this architecture encounters issues of scalability and increased latency, as is commonly observed in centralized systems. The regulatory authority imposes restrictions on the airspace utilization by operators and mandates performance-based authorization for UAVs based on their technological capabilities. Consequently, operators are compelled to act responsibly by submitting a comprehensive flight plan and obtaining the regulator's authorization to occupy designated airspace within a specified geographical area during a specific time slot. Furthermore, the USS must obtain a certification before offering services to operators. We illustrate the centralized architecture based on regulator rules with two UTM initiatives from the literature: the US UTM [36] and the UTM of India [68]. The project of the US UTM was launched in 2016 under the collaboration between the FAA and the **National Aeronautics and Space Administration (NASA)**. They created the **UTM Research Transition Team (RTT)** responsible for the development and implementation of the UTM. In 2017, the project established the **UTM Pilot Program (UPP)** to test the initial UTM capabilities. In 2019, FAA tested with several companies namely the General Electric, Airmap and Uber the UTM implementation. The project went through various implementation steps named **Technical Capability level (TCLs)**. An official document, namely the **concept of operations (conOps)**, is provided by the FAA to give a description of the proposed UTM system, the essential rules and regulations enabling the integration of UAVs in the airspace. The conOps was edited in two versions: ConOps V1.0 in 2018 [4] and ConOps V2.0 in 2020 [36]. The architecture of the US UTM is mainly related to the regulatory and authority roles of the FAA (see Figure 1). The **Flight Information Management System (FIMS)** is considered as the central entity of information sharing. It allows a bidirectional communication to exchange airspace constraint data between the FAA and the UTM. The USS communicates with almost all other stakeholders of the system. It connects the operators with the UTM system and with other supplemental data services. It can

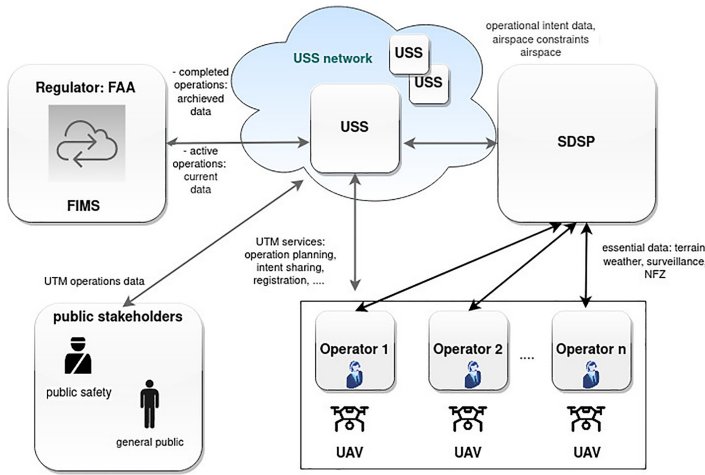


Fig. 1. US-UTM architecture.

archive the operation's data and provide them to the FAA in case of a request. Furthermore, it can support **Remote Identification (RID)** capabilities. The operator shares with the USS the operation intent and real-time information and receives other operators' operations intent, airspace constraints and notifications [36].

There exist additional government initiatives, based on the FAA concept of UTM architecture, including the **UTM in Korea (K-UTM)** [57] and the **UTM in Japan (JUTM)** [58]. Several manufacturers are engaged in partnerships with the FAA and NASA, such as Uber, Skyward, Amazon PrimeAir and Airbus, to develop UTM systems capable of providing essential services to operators for the safe execution of their operations.

The UTM of India is the project of the Indian Ministry of Civil Aviation, launched in 2018 [68]. Similarly to the FIMS, its architecture includes a central entity named the digital sky platform block. It reflects the centralized aspect of the UTM, and it is composed essentially of the digital sky engine. It manages several databases, and it is the single point of truth of the critical information. The digital sky government is the user interface between the UTM system and the governmental institutes. It provides a global airspace situation description to the entities concerned with security, like law enforcement. Whereas, the digital sky UTMSP collaborates continuously with the private UTMSPs. The latter acts as the user interface between the UTM system and the UAV operators to ensure services like registration and airspace authorization (see Figure 2).

3.2.2 Hierarchical Centralized Architecture. In a hierarchical UTM architecture, the airspace hierarchy extends to the UTM system, with separate but complementary UTMs employed for non-segregated and segregated airspace. Nonetheless, this UTM architecture maintains a centralized entity, responsible for disseminating information to all stakeholders. To the best of our knowledge, the UTM of Taiwan is the only initiative emphasizing the hierarchical aspect in the UTM system. Here, the concept of hierarchy is mainly based on dividing the airspace according to the altitude [64]. The part of the airspace where the altitude is less than 400 ft is considered as a non-integrated airspace. It is managed by a **regional UTM (RUTM)** under the control of the local government. On the other hand, the airspace where the altitude is more than 400 ft is an integrated airspace. It is managed by a **national UTM (NUTM)** under the control of the **Civil Aeronautical Administration (CAA)** (see Figure 3). The **Air Navigation Service Provider (ANSP)** intervenes when the UAV exceeds the altitude of 400 ft. It transmits the navigation data from the ATM to

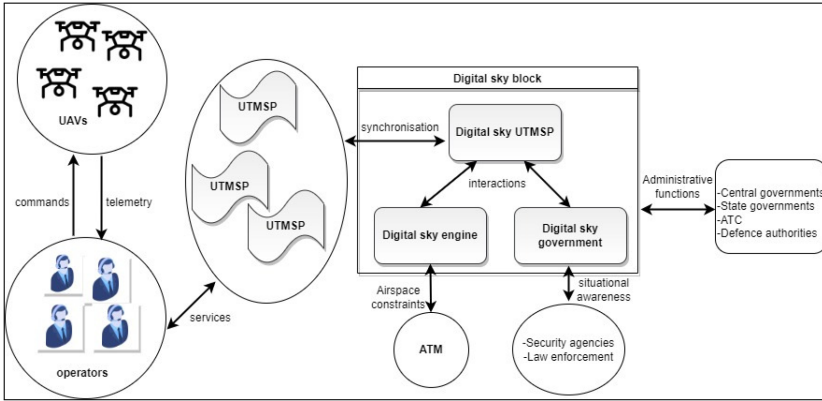


Fig. 2. UTM India architecture.

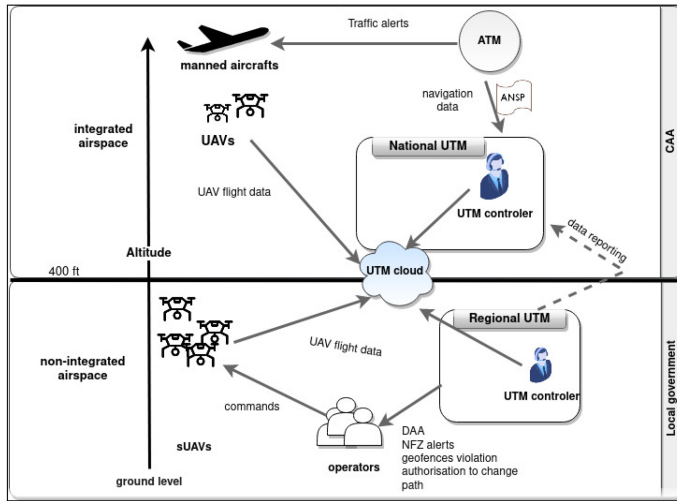


Fig. 3. Hierarchical architecture.

the UTM cloud. Whether the UTM is regional or national, its centralized architecture includes the UTM cloud and a principal UTM server. The role of the UTM cloud is to receive UAV surveillance data. Whereas, the principal UTM server receives and treats data from the cloud. This approach focuses primarily on the surveillance function of the UTM. A communication infrastructure is created to allow UAVs to transmit the position automatically to the UTM cloud in real time. An **Automatic Dependent Surveillance-Broadcast (ADS-B)** like communication technology is imposed to be embedded in the UAVs. The operators bear the responsibility of independently resolving conflicts. Adopting a hierarchical UTM is a good idea to deal with the separation of UAVs with the manned aircraft. It decreases the overhead compared to use a single UTM for managing both parts of the airspace: segregated and not segregated. As previously stated, this responsibility currently rests with operators, which inhibits achieving a high density of UAVs. This task can be shared between the RUTM and NUTM based on altitude. Nonetheless, this arrangement necessitates ongoing collaboration and continuous updating of common information and surveillance data as a prerequisite.

Primary tests are elaborated in Taiwan and performed in an altitude less than 400 ft. It should be noted that this architecture has not been tested in the integrated airspace and under critical conditions (high traffic density).

3.2.3 Centralized Service-oriented Architecture. The **Service Oriented Architecture (SOA)** is a way of reorganizing software applications and infrastructure into a set of interacting services [70]. Generally, the use of a SOA is founded on the simplicity of the system integration, the sharing, and the reuse of services from several providers [76]. However, the services must use common interface standards and protocols to be rapidly incorporated into new applications [65]. In the context of the UTM architecture, the SOA is used by the U-space which is the project of the **Concept of Operation for EuROpean UTM Systems (CORUS)**. It is initiated in 2017 by **SESAR Joint Undertaking (SJU)**, the institutionalized European public-private partnership, and it is yet under development. The implementation of U-space involves four distinct phases, wherein each phase entails the implementation of new services, taking into consideration progressively higher levels of risk and traffic density. A summary of U-space development phases is given in Table 1 that is provided as online supplementary material. It is possible to develop several U-space in one country. The establishment of a U-Space within each member state depends on factors such as traffic density and the characteristics of the region (urban, rural, etc.). Within each U-Space, a single **Common Information Service (CIS)** provider will be designated. It is the central entity providing the critical services and considered as the only point of truth. The regulatory authorities retain direct control on the CIS or delegate it to only one designated service provider. Non-critical services are provided by the operator USSP. It is considered as an interface between the UAVs and operators. An operator USSP must supply four mandatory services, that are the network identification, the traffic information, the geo-awareness, and the UAS flight authorization. The weather information and the conformance monitoring services are optional. They can be provided by the operator USSP or the SDSP [31]. As mentioned before, the SOA has numerous advantages. However, allowing multiple **U-space service providers (USSPs)** to operate simultaneously in the same volume of the airspace makes the connectivity and interoperability between the services and related systems a necessity [65]. The operators are offered a range of services, but they are required to adhere to legal provisions regarding airspace usage. To ensure compliance, operators are obliged to submit their flight plans and intentions to a certified USSP and share this information during the course of their flights. The USSP, in turn, must be flexible and adaptable to meet the specific requirements of the operators while adhering to regulatory guidelines. Several countries have implemented U-space services in accordance with EU requirements and regulations through governmental initiatives in collaboration with private companies. While maintaining a single source of truth for each country, U-space provides private enterprises with the opportunity to develop the CIS. These countries include Croatia, which has established an integrated UTM in collaboration with two companies, D-fend and AMC Portal. The latter has even been chosen as a CIS [8]. Latvia has implemented U-space services through collaboration between the air navigation service provider LGS and the domestic technology innovation company LMT [9]. Switzerland has adopted U-space services through a public-private partnership between the Switzerland's **Federal Office of Civil Aviation (FOCA)** and the **Swiss U-Space Implementation (SUSI)**. This country has even successfully finalized its ConOps [38]. Spain has implemented U-space services through the Spanish air navigation service provider ENAIRE, in collaboration with the Ministry of Defence [25]. France launched the "U-space Together" program which is created by DSN (French Air Navigation Service Provider) in cooperation with UTM Service Providers [35]. Germany provided a commercial UTM system based on U-space in big cities where there is sufficiently high demand. The project is installed with collaboration between the Deutsche Flugsicherung GmbH (the German ANSP) and Telekom [35].

Furthermore, several manufacturers have developed UTMs while adhering to U-space services, including AirHub, DRONIQ, and Indra.

3.2.4 Centralized Architectures based on Specific Cellular Network. The cellular networks are used in the UTMs to ensure necessary communication links between the different stakeholders. The **Long-Term Evolution (LTE)** is used especially for UAVs to benefit from the UTM services using the D2I communication [81]. The UAVs use the LTE in VLL at high density. Tests have demonstrated that due to the UAV uplink interference, communication losses of several seconds can be observed. Hence, surveillance and traffic management problems may overwhelm. LTE is not the only cellular network used for UTM: 2G, 3G and 4G are also tested for UTM services like the identification and tracking [101]. Their efficiency depends on several factors like the number of UAVs and the region size. The UTMs commonly employ multiple technologies to fulfill communication requirements for each service, including cellular networks, radio frequency-based technologies such as ADS-B and IEEE technologies. The architectures presented in this section, on the other hand, utilize a singular cellular network for all services. The UTM architecture in [88] is based mainly on U-space with the integration of 5G communication systems. This effort is supported by the European Union's Horizon 2020 Research and Innovation Program, under 5G!drones project. This system is still under development. 5G technology provides new concepts including the **5G New Radio (NR)** allowing a higher bandwidth. It promotes a very high quality video streaming. A new physical architecture is created to decrease the latency for **Command and Control (C2)** links. Hence, [88] highlights how a communication infrastructure based on 5G technology can support interactions between the different stakeholders and facilitates U-space services. The UTM is an actor in this U-space architecture. It provides federated services and holds all the information necessary to accept or refuse a flight. The proposed 5G infrastructure is managed by the 5G owner, who is responsible for providing all the necessary information about the coverage and the 5G connectivity on a specified airspace volume. As mentioned in [88], the infrastructure is based on the 5G core network, 5G radio base stations and an edge data center. A positive point in this architecture is that instead of considering the network management in U-space as an additional service provided by an SDPS, it will be treated by a whole block named 5G architecture. However, [88] presents just a high-level architecture and there is no actual testing for field performance of 5G platform.

The adoption of 6G network technology in the design of UTM systems is argued by the necessity to manage the airspace for mobility purposes. Several problems surrounding the network coverage, **Beyond Visual Line of Sight (BVLOS)** communication and interference for 4G, 5G and 5G NR promoted the idea of having advanced communication technology. The idea, developed in [86], is to manage the airspace with a multi-layer architecture that includes heterogeneous technologies. The airspace is divided using air, satellite, and terrestrial network to have as the result a multi-layer **Space-Air-Terrestrial (SAT)** architecture: (i) Space design: using **Geostationary Earth Orbit (GEO)** and **Low Earth Orbit (LEO)** satellites, (ii) Areal design: using HAPS, (iii) Terrestrial design: using terrestrial cellular base stations. According to [86], the 6G combined with the RADAR technology gives a very high location accuracy with the centimeter level compared to 4G in which the accuracy is around 10 m and for 5G NR is 1 m. The 6G with the satellite technology assures a connectivity for long distances and real-time control of UAVs where the latency is less than 1 ms, whereas it is around 100 ms in 4G and 10 ms in 5G NR. Likewise, the HAPS with the 6G allows an improved network coverage. As mentioned in the paper, 6G is yet under development in the hope that it will be used in the UTM by 2030.

3.3 Advanced UTM Architecture

Shrestha et al. propose in [86] an advanced architecture of the UTM. It is essentially based on the idea of an autonomous UTM based on 6G/SAT networks. Even more, it is an open source, and

has an architecture based on the cloud. Practically, It is composed of almost the same entities of a basic UTM architecture, with more advanced communication technologies between stakeholders. In fact, the UTM operation system is related to every USS via local cloud USS. A link between **Non-Terrestrial Network (NTN)** satellite with the SDSP which is in turn connected to USS to provide supplemental data about navigation, weather, terrain, obstacles, and surveillance. The UAS operators communicate directly with the USS to benefit from the UTM services (airspace management, dynamic routing, **advanced Detect And Avoid (aDAA)**, monitoring) and autonomous aircraft which can communicate with ground vehicles using access points to share operation intent and real-time information for safety reasons. UAVs can communicate between them using side link with U2U communication. Infrastructure links between NTN satellites and UAVs are performed with **UAV-to-Infrastructure (U2I)** communication. **Machine Learning (ML)** techniques like **Reinforcement Learning (RL)**, **Deep Learning (DL)** and **Convolutional Neural Network (CNN)** are used to make the UTM more intelligent and able to work without any human intervention to regulate the air traffic dynamically and autonomously. [86] indicates an interoperable protocol that will assure a seamless communication between all entities and a database having a distributed architecture to store specific information concerning the operations and UAVs.

3.4 Decentralized Architecture: The Blockchain UTM

A distributed architecture based on the blockchain answers the single point of failure problem. Hence, it requires a limited or no central authority [11]. The blockchain technology is foreseen as the research direction for the decentralized UTM architectures, even for companies. For instance, since 2018, Boeing has invested 32 million dollars to develop a UTM based on blockchain technology and artificial intelligence [34]. Likewise, DEEP AERO, the United Arab Emirates company, announced in 2018, that it planned to build an autonomous, intelligent and decentralized UTM based on AI and blockchain to improve the urban transportation and cargo delivery [39]. In the realm of the UAVs, blockchain technology is regarded as a viable and efficient solution due to its lightweight nature. This makes it well-suited to address the inherent limitations of drones, such as their limited computing capacity, storage, processing memory, and energy resources [12]. The **Internet of Drones (IoD)** concept allows a high-level autonomy of drones located in the cloud. However, different security problems can influence negatively the flight path of the drones that may threaten public safety. Threats can vary from the fake communication messages between UAVs and ground stations caused by attacks like jamming, **Ground Control Station (GCS)** control signals spoofing or man in the middle, to deviated trajectories and collision risk. To mitigate these issues within the UTM, [12] proposes the utilization of a blockchain architecture. The key participants in this framework are as follows: (i) UAVs which are autonomous pilotless unmanned aircraft or remotely piloted by a GCS with various sensors; (ii) ground control stations, responsible for receiving telemetry data and controlling UAVs via C2 messages; (iii) the blockchain network, serving as an immutable distributed database for sharing recorded transactions among network nodes; (iv) the cloud server, employed to overcome computing limitations of drones; and (v) Users, who can access UAV and GCS data for specific missions (refer to Figure 4). To augment the storage capabilities of the blockchain, the researchers employ a distributed and peer-to-peer database called orbitDB, commonly utilized in decentralized and blockchain applications. Furthermore, they leverage the **InterPlanetary File System (IPFS)**, a data transfer protocol, to uphold data immutability in storage [12].

In a similar context, [11] employs a crowd-sensing approach and the blockchain technology to develop their UTM system. The main idea of the paper is to assure the RID, the mission planning and the authentication UTM services. They use a public blockchain with a distributed ledger to manage the operator's access to the airspace. The smart contracts are employed to manage the

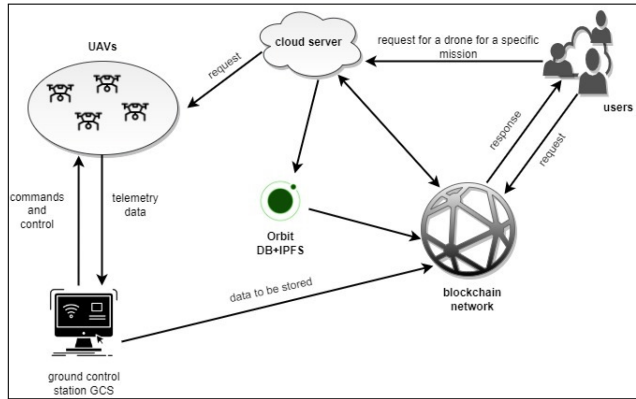


Fig. 4. Blockchain UTM architecture [12].

authorization process. The information stored is about the drone, its operator and its mission. In addition, in each block, we find the information mentioned before in the form of transactions and the states of the smart contracts. Recently, numerous studies tend towards the combination between blockchain technology and **Mobile Crowd Sensing (MCS)** to ensure the decentralization and overstep the limit of the single failure point. In this paper, they employ the MCS to receive remote ID information from the smart devices users. This is considered the basis for the UAV tracking and monitoring. Regarding the UTM itself, they keep the principal actors of the existing classic UTM architectures (UAVs, operators, USSs and the general public using smart devices) and the same concept of the remote ID respecting the FAA regulations. Each operator, aiming to carry out a mission, goes through several steps to be able to benefit from the services of the UTM. A smart contract function is called to accomplish every step. At first, the operator must register its UAV to be assigned an ID and be added in the authority database. Then, he must subscribe to a USS to plan his flight and benefit from the USS services. In fact, the USSs are responsible for scheduling the UAV missions and deconfliction. The operator receives the details of the flight plan in the form of a hash with the **remote ID verification code (RID-VC)** and the remote ID. It must then broadcast the RID-VC and the remote ID throughout the flight. Finally, it will be monitored with the MCS. This initiative has rather a commercial aspect compared to [12]. The participation of the UAV in the airspace is subject to a cost named **refundable compliance deposit (RCD)** and the concepts of penalties and rewards once the mission is completed.

3.5 Monolithic vs Federated UTM Architecture

The UTM deployment plan is another way to classify the UTM architecture. It is defined as the strategy that must be considered during the planning of a UTM in one country, and it depends on the density of the traffic and the types of the UAVs [68]. It concerns the degree of distribution of the services among the USSs. The USS (also named USSP or UTMSP) is an entity that allows the operator to be connected to the UTM system services and with other supplemental data services. The main services that can be provided by a USS are those of registration and identification, operation plan sharing in the pre-flight phase, and the intent sharing in the flight phase to improve the strategic and tactical deconfliction. There are several industry partners that can provide UTM services to the operators including ANRA Technologies, Airmap, AiRXOS and Wing [89]. A UTM architecture is monolithic, if all services are deployed to only one USS in a specified volume of airspace. Conversely, in a federated deployment, non-critical services can be provided by several USSs. The critical services include the exchange of information, data provision and navigation

services [31]. The USS network allows the USSs to collaborate and exchange information like intent data and airspace constraints. The FAA announces that the UTM is not a monolithic system but it is federated allowing to companies providing a set of services that the operator may ask for [36, 89]. U-space, being an SOA, determines the deployment plan on a service-by-service basis. In fact, there are services that can be operated in parallel, while others function more efficiently as a singular state-based entity [31]. For the UTM in India, there are three strategies: (i) The single UTM strategy: A single UTMSP is allocated to a specific airspace and the airspace regions are separated. (ii) The multiple UTM strategy: More than one USS are allowed to manage a specific airspace, and they collaborate and communicate together to provide necessary services to the stakeholders. (iii) The hybrid strategy: allows the two types of strategies described above, a specific airspace can be managed by only one UTMSP or more than one, depending on the type of operation. The hybrid strategy is adopted in the design of the India UTM.

We conclude through this section that the degree of centralization or decentralization in an architecture predominantly relies on the decision to distribute or not distribute. This choice is influenced by several factors, including the security, safety, capacity, and interoperability. Advanced UTM is primarily based on advanced communication technologies and, notably, the AI to automate UTM tasks. The challenge of integrating AI within the UTM framework involves developing robust, reliable, and efficient AI systems capable of handling the complexities of real-world scenarios, ensuring safety and collision avoidance, scaling with the increasing number of drones, adapting to dynamic environments, and addressing privacy and security concerns. We provide a comprehensive summary of the architectural features, consolidating them in a comparative table (see Table 4). The decision to distribute UTM services to a single USS in a monolithic manner or to multiple USSs for a more federated approach depends on factors such as cost, complexity, and interoperability. The most developed UTMs adopt a federated approach to provide manufacturers with opportunities to contribute to the development of services for operators while adhering to a set of rules, standards, and regulations imposed by the regulator entity.

4 UTM SERVICES

Each UTM system provides a set of services aiming to ensure the security, safety and successful completion of the UAV mission (operation). The UTM participating entities provide a set of services. In [33], the authors summarize the services that each entity can offer to other entities. The **UTM Service Provider (UTMSP)** is responsible for providing essential UTM services, including communication and navigation services, as well as network identification. The supporting or value-added service provider supplies supplementary services, such as weather, terrain, and obstacle information, to all other entities. Meanwhile, the Common Information Services provider facilitates the exchange of information among UTM stakeholders. The regulator is the civil or federal Aviation authority in a country. It establishes the constraints of use of the airspace. In the same context, [68] groups the services into UAS services, UTMSPs services, SDSPs services and digital sky engine services. (More details concerning the classification of the UTM services according to concerning entities are provided in Table 2 that is provided as online supplementary material). Otherwise, every mission undergoes a series of stages that delineate its lifecycle (Service classification according to the operation phase is given in Table 3 that is provided in online supplementary material). Before planning its flight, the operator must have the approval and the licencing from the regulator to use a part of the airspace during a period of time. Once the flight plan is prepared, the operator with the USS gather additional information pertaining to weather, terrain, and obstacles and employs a strategic flight management. During the flight, the operators share the real time flight information as a means of deconfliction. In the in-flight phase, a tactical deconfliction is employed based on the tracking and monitoring services. After the mission is

Table 4. UTM Architectures Comparison

Reference	UTM designation	Architecture category	Central entity	Main features	Advantages	Issues and challenges
[36]	US UTM	centralized based on the regulator rules	FIMS	-FAA: regulatory and authority roles -FIMS: data exchange between FAA systems and UTM participants -USS: provide services to support the UAS community	-Efficient for the integration of sUAS in the VLL airspace -A single entity ensuring information sharing is more cost efficient -Safe and accurate shared information	-no adequate for the users of higher altitude -The single point of failure issue
[68]	UTM of India	centralized based on the regulator rules	digital sky engine	-digital sky block: regulatory role, enables collaboration between stakeholders -UTMSP: deconfliction, tracking, conformance monitoring, constraint and contingency management	-Safety and security aspects are ensured -Open access -Clear and direct governmental oversight	-The single point of failure issue -Scalability problem -Efficiency in complex operation
[31]	U-space	centralized service-oriented	CIS provider (principal DTM)	-CIS provider: provide critical service -USSP operator: provide services to drone operators	-Simplicity of the system integration, the sharing and the reuse of services from several providers	-The single point of failure -Necessity of common standards and protocols to integrate services -Cybersecurity challenges
[26]	UTM based on in-flight services	centralized service-oriented	U-space service manager	-U-space service manager: communicates with all the actors of the system -database: manage information about UAS operations and active geofences	-The same as U-space	-Too many modules to track and monitor UAV flights
[88]	5G U-space	centralized based on cellular network	CIS	-The customer block: collaboration between the end user and the business provider -The U-space block: UTM services for operators -5G infrastructure block: connectivity and coverage information	-Advanced communication infrastructure -The 5G infrastructure is a whole block -5G: less latency and higher bandwidth	5G technology limits: -Availability problem in non-urban areas -Interference problem -Limited coverage for PAVs and UAVs in 3D environment
[86]	6G UTM	centralized based on cellular network	FIMS	-multi-layer SAT architecture Space design, aerial design, terrestrial design -communication based on 6G technology	SAT architecture: -high position accuracy -long distance connectivity -real time-control of UAVs: latency less than 1ms -enhanced network coverage	-The lack of the system evaluation in real conditions -Air transportation challenge
[64]	UTM of Taiwan	centralized hierarchical	principal UTM server	-RUTM : manage the non integrated airspace -NUTM: manage the integrated airspace -UTM server: receives and treats data from the cloud -UTM cloud: receives UAV surveillance data -Blockchain network: maintain an immutable distributed database and share recorded transactions between network nodes -OrbitDB+IPFS: improve the storage capacity of the blockchain -Cloud server: overcome drone computing limitations	-The overhead is decreased compared to using a single UTM for managing both parts of the airspace: segregated and not segregated -No single point of failure -No central authority requirement -Blockchain technology: secure transactions -High level autonomy of drones -Reduce security problem	-Not tested in non segregated airspace -Efficiency in complex operations and critical conditions
[12]	Blockchain UTM	decentralized	no central entity	-Public blockchain+ distributed ledger: manage the access of operators to the airspace -Smart contracts: manage the authorization process -Mobile crowd sensing: receive remote ID information from users of smart devices	-limited requirement for central authority -Mobile crowd sensing: ensure decentralization -Smart contracts: no need to intermediary or legal system	-Consensus protocols are hard to develop -Latency issue in decentralized systems
[11]	crowd-sensing blockchain UTM	decentralized	no central entity			-Same limits as [12]

completed, a recording and a report of the mission are generated. Subsequently, we delve deeper into the fundamental and prevalent services provided by a UTM.

4.1 Registration and Identification Services

The registration is the first service provided by the UTM. It allows both the operators and UAVs to be registered, to benefit from the other UTM services. The UAS registration is managed by a UAS registration system which allows the access to registration information by involved entities such as the law enforcement and the authorized public [40]. Since the number of drones flying in the sky is becoming more and more important, the identification is essential for the traceability and the tracking of the UAVs. It is a means for determining the intruders and the UAVs not authorized to use the airspace. Different methods for the UAV identification have been used since the creation of the first UTM systems until now, notably, the **flight alarm (FLARM)**, the **Automatic Dependent Surveillance-Broadcast (ADS-B)** and recently the **Remote identification (RID)**. At first, ADS-B and FLARM were not created for UAS identification. The ADS-B is a surveillance technology intended to be used for the manned aircraft. It allows broadcasting periodically messages to be received by the nearby aircraft. The messages provide information about the identification of the aircraft, the current position, the altitude, and the velocity. They can be transmitted every 2 seconds. ADS-B is a good surveillance technology for manned aircraft, essentially in areas not covered by radars [73]. However, the use of ADS-B for the UAS identification is encountered mainly by security, interference, and congestion problems [78]. FLARM in its primary version was a collision avoidance technology for light aviation. In 2018, the FLARM UAS electronic ID was released to provide UAS an electronic identification and a 3D position broadcasting. The messages are transmitted every 3 seconds. It allows a more secured mechanism of identification than ADS-B,

Table 5. RID Methods Comparison

RID method	Broadcast RID	Network RID
Transmission	-one way direct transmission mode -one to many communication	-through the Internet -one to one communication
Destination	all local devices in the range of the broadcast	a qualified USS
Technologies	-Wi-fi -Bluetooth	cellular technologies: LTE, 5G
Advantages	-secure -safe	-facilitates for the USS information sharing -facilitates the access to RID data from law enforcement and general public -allows BVLOS operations
Limits	-restricted range -more energy consuming -collision problem when using a predefined frequency band	-limited in the area not covered by the network -security and privacy issues

but it suffers from packet collision and transmission delay issues [78]. Since 2019, almost all the **civil aviation authorities (CAA)** have required the use of RID technologies to identify the UAVs and their operators [28]. The RID message includes identification and location elements. The three identification technologies are message-based but they differ from the type, size, format and frequency. For more details, we present in Table 4 that is provided as online supplementary material, a comparison between the UAS identification methods. In [68], other elements in a RID message are mentioned as the latitude, the longitude, the intent information and the emergency status of the UAS such as lost link. There are two methods for RID transmission, which are the broadcast and the network RID. The broadcast RID (direct broadcast), is characterized by a one way direct transmission mode with no specific destination. When the RID signal is emitted, all the mobile devices that are in the range of the broadcast can receive it. The broadcast RID is implemented by Bluetooth (Bluetooth 4.x and Bluetooth 5.x) or Wi-Fi (Wi-Fi NAN at 2.4 GHz and Wi-Fi NAN at 5 GHz) technologies [27]. It is limited due to the restricted range of these technologies, in addition to interference, privacy, and energy consuming problems for UAVs [68]. [44] proposes to use the LTE D2D and the side link communication in the direct broadcast. The two technologies are introduced by the 3rd **Generation Partnership Project (3GPP)**. They are useful in ensuring communication without network for devices that are not covered by the cellular network. The network RID, is to transmit the RID information from the UAS to an USS over Internet. The USS must be qualified by the regulator to provide RID services [36]. The network RID is implemented especially with cellular technologies and will be limited in the areas that are not covered by the network. In Table 5, we provide a comparison between the broadcast and network RID. The FAA stopped using ADS-B and announced that it will employ the broadcast RID rather than the network RID for security reasons [36]. For the UTM of India, the choice of the RID method depends on the type of the UAV and the application's category for which it will be used. In fact, the network RID is adopted for UAVs working for the law enforcement agencies in fire and disaster management. Whereas, the UAVs used for defense often implement the direct RID for security purposes, but this does not exclude that they could implement the network RID. It is possible that the large UAVs, which the weight is at least 150 kg and size is between 20 m and 50 m, used for defense being equipped with other means of identification like ADS-B and SSR transponder [68]. The need for a standardized broadcast transmission leads to the development of protocols based on the nation's regulations. The ASTM F3411-19 inspired from the US regulations, the prEN 4709-002 inspired from the European regulations and the **Remote Identification Protocol (DRIP)** developed by the **Internet Engineering Task Force (IETF)** DRIP group to facilitate the RID functions standard.

For the decentralized blockchain UTM, the identification of nodes is carried out using the pseudonymous addresses. The identification system allows the registration of the UAVs and their

tracking data [12]. This does not prevent that some blockchain UTM use the existing RID approach, like in [11].

4.2 Tracking, Surveillance and Monitoring Services

The tracking service provides information about the airspace situation (the situational awareness). It provides the UAS current positions and predicts future trajectories to concerned stakeholders like the ANSP and the USSP [26, 68, 96]. The prediction of the future trajectory is based on the current position of the UAV, its velocity and its flight plan [26]. In the case of incorrect or missing data from the UAV, the service sends a notification message to the concerned operator. For U-space, the tracking is among the initial services to be developed in the U2 phase (we present in Table 1, that is provided as online supplementary material, the U-space development phases). It depends on several other services and sub-services. In fact, the position report sub-service is responsible to send at regular time intervals position reports to the tracking service, which in turn, generates tracks to be used by the monitoring service [31]. Nevertheless, a main issue arises when the UTM deals with non-cooperative drones. They may cause real risks of danger as long as they do not transmit RID signals or position reports. The **drone detection system (DDS)** is the technology that may be a solution to detect uncooperative drones at a range of 500 to 5,000 meters [31]. However, it is limited since UTMs are generally regional or national. The proposed system in [24] can support uncooperative UAVs by adding tracks of non-cooperative sensors like radars, infrared and cameras. Mathematically, the prediction of the future UAV positions is implemented by statistical filters. The tracking module generates tracks and estimates the UAV future positions by merging the information deriving from several sources. In [26], the Kalman filter treats the noisy and inaccurate telemetry measures provided by the ADS-B transceivers before generating predicted trajectories. In [24], the tracker implements two Bayesian filters. The horizontal filter is used to predict and filter the x and y positions. The vertical filter is used for z position and velocity. Even in U-space [31], the building of tracks is carried out according to a statistical process based on the flight plans of the UAVs. The filters can alleviate the data errors and imperfection, but are enabled to eliminate them totally. The monitoring service is linked by the information flows with the operators, USSP and authorities to give a view of the state of the airspace [31]. The monitoring information is collected from the tracking service and combined with static and dynamic geofences and obstacle information. The main function is to detect the eventual conflicts that are the following [26]: (i) a conflict between the UAS trajectory and a geofence; (ii) a deviation of the UAV from its operational volume that it must respect throughout its flight; (iii) a loss of minimum separation between two UAVs. In [26], the monitoring service regularly accesses a database to read information about the UAS active operations, the static geofences that are created in the pre-flight phase and the dynamic geofences created during the flight. Geometric methods are mainly used in their approach to detect conflicts. The UTM in [66], is based on a self UAV monitoring through an embedded flight controller. The UTM receives alerts from the UAV when it is in danger. But, that does not prevent from being overseen by an UTM director.

In a similar context, the surveillance service communicates with the tracking service and other information sources. As mentioned in [31], these sources may be primarily the radar and drone detection similar systems, cellular telephony information and similar, motion capture systems, ATM aircraft surveillance systems and neighboring drone trackers. [36] summarizes the means of collecting information for a specific geographical area by the terrestrial, the airborne and satellite surveillance. The responsibility of providing surveillance services can be attributed to a surveillance service provider [31]. The technologies used for UAS identification are also used to ensure surveillance, for instance the ADS-B and FLARM. [73] describes in detail the different technologies and protocols used in collaborative surveillance in UTM systems.

4.3 Conflict Management Service

The objective of the conflict management is to avoid physical collisions between the UAVs and static or mobile obstacles namely the buildings and structures, people, animals, manned aircraft and other UAVs. Almost all the existing studies in the literature, present the conflict management as a process of three layers [31, 36, 68, 96, 97] namely the strategic deconfliction, the tactical deconfliction and the collision avoidance. The strategic deconfliction (mission scheduling) is done before the UAV flight. It includes the actions of flight planning, geofencing and airspace design. The flight planning allows the operators to submit their flight intent based on precise information about weather, terrain, and airspace constraints. The operators must maintain a continuous communication with the USS to have a real time and a detailed view of the airspace [36]. The sharing of the plan intent between operators across the USSs significantly reduces the risk of collision between UAVs. The strategic deconfliction is triggered in two cases: if a new flight plan of an operation is submitted or if the plan of an operation is changed [31]. The detection of a conflict probability or a loss of separation between UAVs is based on the investigation of the 4D trajectories that are predicated from the UAV operation plan [31]. The urban spaces present a real challenge in avoiding collisions compared to sparsely populated and rural areas. In reality, they are characterized by a significant human population densities and a well-developed infrastructure and economic activities. Which means that air risk and ground risk are significant. In this context, [66] has defined three fundamental building blocks of urban airspace that the UAVs must respect in their flight plans. [66] uses the safety tube concept for deconfliction. It is defined as a spatial or temporal safety margin or tube required for the drones to handle inconsistencies due to delays. It must be active 15 seconds before the execution of the plan flight and 60 seconds after with a radius of 10 meters. The role of the USSs is to receive the UAVs paths' requests that are accepted if they do not present any conflict with the **no-fly zones (NFZ)** and **prior permission zones (PPZ)** or with existing UAVs trajectories. The tactical deconfliction (separation provision, remaining **well clear (WC)**) encompasses all the actions that must be performed during the UAS flight to resolve conflicts that may occur despite the efforts made in the strategic deconfliction. As mentioned in [31], in a non-segregated controlled airspace, ATC is responsible for tracking aircraft and predicting their trajectories by using radars. The TD is directly linked to the tracking service to have the positions of the UAVs operating in the airspace in real time, which is a necessary condition to predict any possible conflict. In [26], the role of the TD module is to generate alternative flight plans to solve a conflictual situation. The decision-making concerning the solution to be adopted is attributed to the **emergency management (EM)** module. Its decision is based on a cost and a riskiness level computed by the TD module and reported to him with each proposed flight plan. The TD can be faced with two possible conflict situations, and for each situation it applies a specific approach. In fact, the conflict may involve multiple UASs or a single one. When multiple UAV flight plans conflict due to a loss of separation, the solution is to modify flight plans by using a geometric approach named the repulsive forces (see [26] for more details). Whereas, if a UAS is out of its operational volume or is conflicting with a geofence the solution is to use a heuristic path planner through the A* algorithm [26]. In fact, if the UAV encounters a geofence, the path planner generates an alternative trajectory in order to circumvent it. Whereas, if the UAS is already in a geofence, it comes out by an escape point, which is the first point on the geofence's border closest to the UAS. [66] develops almost the same process as [26] for the tactical deconfliction. Except that in [66], the triggering of this function is done by the UAV itself or by the UTM with a secondary status monitoring system for the UAV. It is mentioned that for a delay of 45 seconds, the UAV alerts the UTM. The collision avoidance is the last layer of the conflict management. It is triggered in case of failure of the first two deconfliction functions. The only way to avoid the collision is to

make maneuvers by the UAV itself. [97] gives an overview of the existing approaches dealing with conflict management. The time-based approach presents the reference point as the expected time of the potential collision. Timing boundaries are applied to conflict management layers. The limit of the time-based approach is that some parameters that can influence the time-to-collision, are not well specified by the approach. In fact, it is not well clear how to treat vertical separation between UAVs and which velocity must be used (instantaneous, planned, or maximum) to estimate the time-to-collision. In the distance-based approach, the drone is considered as the center, with two cylinders representing the safety area. The boundaries depend on the deployment environments. In urban areas, the boundaries must be smaller than boundaries in suburban environment with wide-scale drone deployment. The limit of the distance-based approach is not considering the velocity of the UAV intruder. Wireless technologies are used in the conflict management to send the position of the UAV and ensure the communication between the UAV and the ground station or between the UAVs themselves in the space. The strategic deconfliction requires the technologies that have a ground infrastructure like LTE and **Long Range (LoRa)**. However, the tactical deconfliction needs the broadcasting technologies namely **Automatic Packet Reporting System (APRS)** and Wi-Fi SSID. For more details, the reader can see [97].

5 AIRSPACE STRUCTURE FOR UTMS

The airspace structure is to organize and manage the airspace for safe air traffic operations. The manned aircraft are the primary users of the airspace. So that, it was structured and organized to manage manned aircraft disregarding the UAVs. The ICAO airspace classification is the most common airspace organization and it is used by many countries. The UAVs utilize the VLL which is the region of airspace with the lowest frequency of manned aircraft activity. Some efforts are made to create a specific airspace organization for UAVs, like the Corus airspace management for U-space. In addition to the ICAO classes, it adds three volumes based on specific factors that we will detail in Section 5.2. The multilayer airspace management allows a more distributed UTM management. In this section, we discuss these different airspace structures.

5.1 The ICAO Airspace Classification

The ICAO has divided the airspace into seven classes (see Table 6). The A, B, C, D and E classes are considered as controlled airspace. They are allowed to benefit from the ATC separation services. In controlled airspace, the communication with the radio contact is obligatory except for the class E. Whereas, the G and F classes are considered as uncontrolled airspace in which the ATC does not provide any monitoring or control services [63]. This leads the pilots of the manned aircraft relying on themselves to avoid the other users of the airspace while following either the **instrument flight rules (IFR)** or **visual flight rules (VFR)**. In the national India UTM policy discussion paper [68], authors propose by analogy to the ICAO airspace classification an airspace classification for UTM airspace. They argue their choice by the issue of the overlapping responsibilities between ATM and UTM service providers. To this end, the U-airspace which refers to the unmanned airspace will be divided into sub-categories U1, U2 and U3. For each class, the type of the service (**Unmanned Flight Rules (UFR)** for UAVs, VFR and IFR for manned aircraft) is precise. The separation services are provided if the participation of the UAVs and manned aircraft is mandatory and if the UTM clearance is executive or only advisory. For each sub-category, UAVs must respect both the UTM and the ATM regulations. While trying as much as possible to not disturb and avoid manned aircraft in their flights, the UTM uses the VLL airspace for operators to perform their operations. The VLL is defined for the most countries as the part of the airspace below the altitude of 500 ft **above the ground level (AGL)**. Yet, depending on each country regulations, some use lower values, and others higher than 500 ft ranging from 400 ft ([36, 64]), 700 ft

Table 6. ICAO Airspace Structure

	Class	Characteristics
Controlled Airspace	A	
	B	- Radio contact with ATC: is mandatory
	C	- Two way radio communication
	D	- Air controllers: a constantly and complete overview of the airspace traffic
	E	- Radio contact with ATC: is not mandatory - Air controllers: some aircraft are unknown to ATC
Uncontrolled Airspace	F	- No separation service offered by ATC
	G	- Pilots manage the airspace by following air rules

([31, 62]) to 1,000 ft ([68]). For more details, the reader can find in [98] an exploration of the maximum flying height for different countries. There are frequently some misconceptions regarding which class the VLL is concerned with. The VLL is usually confused with the uncontrolled class G. However, it is not limited to a specific class: this part of the airspace can be either controlled or uncontrolled. In addition, the VLL is not completely deprived of manned aircraft. It is true that manned aircraft rarely fly in an altitude less than 500 ft for safety issues, but they can use the VLL for take-off and landing or for military and emergency operations (helicopters). This also does not prevent the use of the UAVs in some applications intended specifically for the controlled airspace, including conducting an inspection of the runway to identify any potential damages, employing methods to discourage the presence of birds, and enhancing prevailing weather conditions [83]. It should be noted that we have not found in the literature a great effort to treat the UAV flights in controlled airspace. Concerning the national initiatives, in [36], the **Low Altitude Authorization and Notification Capability (LAANC)**, which is a collaboration between FAA and the industry, gives authorization to pilots who want to perform operations in the controlled airspace through the USSs. Whereas, in [31], the ANSP is in charge of evaluating and guiding operations in the controlled airspace. Furthermore, [83] presents a solution for visibility and connectivity in mixed ATM/UTM contexts, especially for operations inside or around the airports. It proposes to embed in the UAVs a **Multi-Mode-Transceiver (MMT)** composed of a certified A/C/S transponder supporting ADS-B with a cellular interface to communicate with the ATC. As explained in the paper, the MMT facilitates the visibility of the UAV for both the ATM and UTM. Once detected, the UAV must obtain ATC clearance before coming in the controlled airspace.

5.2 The Corus Airspace Management for U-space

U-space is not limited to the ICAO airspace classification. It divided the airspace based on specific factors of which the more important ones are the expected number of the drone flights, ground risk, air risk, security, and safety factors [21, 31]. Here, the VLL is divided into three volumes namely X, Y and Z (with two parts: Za and Zu). They belong all to the uncontrolled airspace except for the Za part which is under the ATC control. Each volume differs from others by the U-space offered services, the types of operations and the access requirements [21, 31].

The X volume is the part of the airspace with the lowest ground and air risk. It does not need specific requirements on the operator, pilot, or drone. However, the pilot is the responsible for keeping well clear as long as no separation service is offered. The Y volume is characterized by higher air and ground risks. Specific requirements are needed for access to Y especially including an approved operation plan and a UAV capability of position reporting. For this case, U-space ensures only the strategic deconfliction. Whereas, the Z volume presents the highest air and ground risks. Y and Z have the same access requirements. In Z, there are two parts: the Za which belongs to the controlled airspace in which the ATC ensures the strategic and tactical separation and the part Zu which benefits from the U-space separation services. Based on the information from [31], we summarize all characteristics of X, Y and Z volumes in Table 7.

Table 7. Airspace Structure for U-space

Volume	Risks	Access requirements	Operation type	Strategic separation	Services Tactical separation	Tracking
X	- low air risk: few flights - low ground risk: low population density	- Basic requirements on the operator, drone and pilot - Safety is the responsibility of the pilot	- VLOS - Open operations - Other types: air risk mitigation requirements	×	×	×
Y	- higher ground risk: densely populated areas - higher air risk: important air traffic	- An approved operation plan - Specific technical requirements: - position reporting capability, - a remote piloting station connected to U-Space	- VLOS - BVLOS - Open operations (respect of access requirements) - Specific operations - Certified operations	✓	×	✓
Z Za Zu	- higher density operations than Y - highest ground risk : urban areas	- An approved operation plan - Specific technical requirements: - position reporting capability, - a remote piloting station connected to U-Space	- BVLOS - Open operations (respect of access requirements) - Specific operations - Certified operations - Automated flights	✓	✓	✓

✓ : the service is offered, × : the service is not offered.

5.3 The Multilayer Airspace Management

Here, the focus is done only on the uncontrolled class G. The proposal of [62], is to divide the class G into layers (horizontal segments) with different altitudes to create a distributed UTM system. The separation between layers takes into consideration the needed volume for the largest UAV to stay within its segment. This is called the **containment limit (CL)**. Each layer represents an obstacle free space for the UAVs and is composed of nodes and airways which are corridors connecting nodes horizontally, vertically or in diagonals [62]. This approach allows for UAVs three flight modes: lateral flight, vertical flight and hovering, with the possibility to change the mode at the nodes. What is interesting in all that, is the possibility of flying without direct communication between UAVs and the UTM. The advantage is to reduce the latency and solve the loss of communication problem between the UAVs and the UTM. In addition, following the rules of the airway (maximum velocity, maximum traffic capacity) and ad-hoc communication, the UAVs can make local routing decisions in emergency cases.

6 COMMUNICATION IN UTMS

The UTM is based on information flow exchange between all participating entities, making the communication links crucial to conduct command and collision avoidance functions. The needs in communication links differ according to the level of autonomy and the extent between the operator and the UAV, for both VLOS and BVLOS cases. The communication technologies used for UTMs must guarantee a real-time communication, a wide coverage area, a secure networking and a very low latency [87]. In this section, we present the different communication links categories and technologies used in the UTM.

6.1 Command and Control Data Link Communication

The **Command and Control (C2)** communication, or referred to as the **Control and Non-Payload Communication (CNPC)** in some papers ([46, 47]), is necessary to ensure the control of the UAVs safely and securely. In the C2 communication, a UAS may include the UAV and its controller or the UAV, the **Ground Control Station (GCS)**, and the **Ground Radio Station (GRS)** [46]. The GRS ensures the communication between the UAV and the GCS through radio signals. The C2 communication can be established between the controller and the UAV or between the UTM and the UAV. During the flight, the UTM maintains a continuous link with the UAV to transmit the path information and the flight clearance. According to the 3GPP TS 22.125 [1], we distinguish

three control paths: the direct C2 communication, the network-Assisted C2 communication and the UTM-Navigated C2 communication. After being registered, the direct C2 communication allows the UAV to directly communicate with the controller using a radio resource. Whereas, in the network-Assisted C2 communication, they can communicate through the network with a unicast link. The UTM-Navigated C2 communication allows the UTM to intervene in emergency events when the UAV is approaching from a no drone zone or in potential security threats. The information transmitted in a control message differ according to the mode applied. Specific values of **key performance indicators (KPI)** are associated with each C2 mode. Based on the 3GPP report, we give in Table 5, that is provided as online supplementary material, a summary of the C2 control paths and modes with the associated KPIs. In C2 links, critical messages are exchanged (telemetry and control messages), therein, they require a high level of reliability and low latency. In a centralized UTM, the latency of delivering route modification information received from a UTM to a UAS must be less than 500 ms [1]. However, as a non-payload communication, they don't require a high level of data rates. The data rate for VLOS operations is less than for BVLOS. Concerning the BVLOS operations, the C2 communication link sent by the UAV may include a video to aid the UAV control. We present in Table 6 that is provided as online supplementary material, the KPIs for video used to aid the UAV control for the VLOS and BVLOS operations. All the values are given according to the 3GPP TS 22.125 [1] and they concern the data rate, the reliability and the end-to-end latency. Several technologies are used to ensure C2 communication. They can vary from terrestrial cellular networks to satellite connectivity. In fact, if the operation is VLOS, the communication is done through a direct radio link. A BVLOS communication relies on the use of a satellite connection [47] or on more than one connection with redundant C2 links [3]. For the decentralized blockchain UTM systems, the GCS receives data from the UAV and then returns the control commands using the cloud through the Internet. Here, the security aspect is very important. When the UAVs are connected to the Internet, the data flow is surely exposed to attacks. Threats can vary from fake communication messages between the UAVs and ground stations caused by attacks like jamming, GCS control signals spoofing or man in the middle, to deviated trajectories and collision risk. For a more exhaustive study, the reader can see [12].

6.2 Air-to-Air Data Link Communication

The **Air-to-Air (A2A)** links concern both the UAV-to-manned aircraft and **UAV-to-UAV (U2U)** communications. The UAV and the manned aircraft must exchange positional information to avoid collision, especially when the UAV is inside the controlled airspace or around the airports [4]. Ensuring both high visibility and a dependable communication link presents a substantial challenge in such circumstances [83]. The radio technology is used to allow the two-way radio communication with ATC. The U2U communication is especially used in the decentralized UTM to exchange the essential data necessary for the local decision-making in collision avoidance. A multitude of technologies can be used in the A2A communication, but there is not yet a specific standard for this type of communication link. **Flying ad-hoc networks (FANETS)** are candidates for the A2A communication. They provide a direct communication between UAVs and allow transmitting information without needing of a ground system [19]. FANET can also be used for the U2I communication. In the context of the decentralized UTM, the 3GPP sets requirements concerning the U2U communication for identification and separation purposes. An UAV shall be able to transmit at a frequency of 10 messages/s with an end-to-end latency of at most 100 ms. For identification, the UAV must have the ability to locally broadcast its identity with a rate of at least once per second, to a range of 500 m (we present in Table 7, that is provided as online supplementary material, the UAV-to-UAV local broadcast communication for decentralized UTMs requirements based on the 3GPP TS 22.125 report). In the same context, the IEEE 802.11p used for ground vehicles, is being a

candidate for A2A communication. For more details, a summary table is provided in [61] comparing the technologies available for the UAVs based on several criteria including the latency, the range, the rate, and the frequency. Several research papers recommend using the cellular networks in the different UTM communication links [43, 44, 60, 82] to cite a few. They are especially employed for the C2 communication, the network RID and BVLOS links. The short range communication technologies, including the Wi-Fi and the Bluetooth, are used especially in the conflict management, the V2V communication and in the broadcast RID. The adoption of the cellular networks is argued by their mature architecture ready for use, their effective-cost and their availability [43]. Since the majority of the UTM systems proposed in the literature are not yet tested, consequently a few real testbeds are done for the communication technologies in the UTM system. We present in Table 8, a comprehensive study of all technologies used for the UTM based on the papers found in the literature. We explain for each technology the use case for which it is applied in the UTM, the advantages of the technology and limits that are founded.

7 DECISION MAKING IN UTM SYSTEMS

While in flight, a UAV may either be exposed to hazards or present a potential danger. The level of risk associated with the UAV can vary, resulting in varying degrees of impact on its immediate environment. In some cases, this could lead to incidents or accidents with the potential for human fatalities [14]. Therefore, effective decision-making within UTM systems is crucial especially in off-nominal scenarios and emergencies [91]. In this section, our focus is on the decision-making process within a UTM in general. We explore how this process is executed, considering both centralized and distributed approaches.

7.1 Decision Making in Emergencies

According to [14, 91], hazards can arise from the following factors: (i) Technical failures, such as the loss of A2G or **Vehicle-to-Everything (V2X)** communication links, **Global Positioning System (GPS)** malfunctions, power, camera, or engine failures. (ii) Human-related failures, which stem from performance issues of human stakeholders leading to erroneous decision-making. (iii) Imperfect or inaccurate data and data-related issues caused by cyber-attacks. (iv) Infrastructure-related problems that can impact take-off or landing. (v) Environmental events, such as adverse weather conditions. Emergency and off-nominal situations require immediate action and timely decision-making. The UAV has the option to either halt the mission promptly or alter its flight path. To address such contingencies, NASA has developed a decision-making module as part of the NASA SAFE50 project [18]. Onboard decision-making is employed to minimize reliance on air-to-ground dependencies. A **Finite State Machine (FSM)** is implemented, incorporating each contingency as a distinct state [14, 56]. The utilization of the FSM improves the quality of decision-making and reduces response time. In another study, [80] proposes a decision-making framework for UTM systems, centered on the identification and prediction of risk factors. The risk-factors can arise from uncertainties or the vehicle performance degradation. For each UAV trajectory is assigned a degree of safety: safe, off-nominal but safe, abort and the return to base is safe, abort but to return to the base is unsafe. The decision framework is based on a computational architecture that includes three functions: the risk-factor prediction, the decision and trajectory generator. The three functions can be implemented totally on-board the vehicle. This solution allows a reduced dependence and requirements on communication. In fact, on board decision-making process serves to effectively reduce the necessity for ongoing data transmission between the UTM and UAVs, a factor of considerable significance, particularly in scenarios involving communication disruptions. The capacity for local decision-making leads to a decreased occurrence of data transmission in terms of frequency and volume, thereby resulting in a reduction of required bandwidth and

Table 8. Communication Technologies for the UTM

Communication technology	Use cases in UTM	Network scenario	Advantages	Issues	Ref.
Ad-hoc networks					
FANET	-FANET-based UTM system using the WAVE protocol -security, collision avoidance	-A2A -A2G	-availability -to manage status and location of all UAVs	-risk of loss of messages	[71]
Cellular networks					
LTE	-conflict resolution -surveillance -awareness messages	D2I	-capable to provide connectivity for drone in VLL -low-cost deployment	-not efficient in a high density of drones -not suitable in a high message delivery ratios:message drops	[82]
	-strategic deconfliction	A2G	-reliable coverage in large areas	-inter-cell interference problem nm	[97]
4G	-surveillance -DAA: communication between the pilot and the UTM controller	A2G	-efficient as a technology for DAA	-limited altitude -lack of tests	[64]
	-communication between pilot and USS: clearance, updates	-A2G -C2	-allow connectivity for BVLOS operations -doesn't require new architecture	-availability problem in rural environment	[50]
	-to test C2 links in TCL3 of the US UTM	-C2	-low cost -reliability	-dropped links in sudden high demand -C2 link failure -cyber security issues -lack of short-distance direct communication between UAVs	[60]
5G	- using 5G in large scale UAV services	-C2	-network slicing -Service based architecture in 5G core	-limited coverage -non-existent or weak cellular signal in remote areas	[92]
	-control, tracking and communication with remote applications and services	-A2G	-low end-to-end latency -URLLC services -MEC-NFV management and orchestration with UTM system -wide area, limitless connectivity	-a very little number of drones from 5 to 25 UAVs	[23]
6G	-surveillance and monitoring -deconfliction -emergency management	-U2I -U2U -U2G	-cost-effective -high precision positioning -integration of NTN	-the system is not tested	[86]
IEEE technologies					
Wi-Fi	-remote identification	A2G	-hardware availability -cost-efficiency	-short range	[36]
	-real-time information sharing	U2U	-high availability -high bandwidth -no need to establish a connection	-interference issue for UAV environments -energy cost problem	[95]
Wi-Fi SSID	-tactical deconfliction	-U2U -U2G	to a Wi-Fi access point -UAVs exchange coordinates directly -no need of ground infrastructure	-on the ground, no reception at more than 550m	[97]
Bluetooth	-remote identification	A2G	-low cost	-limited range	[36] [31]
X-Bee	-transmission of surveillance data	A2G	-can reach longer distances than Wi-Fi	-lack of tests	[64]
Satellite networks					
LEO, GEO	-BVLOS operations	A2G	-high capacity -robust communication -real time and reliable connectivity over satellites	-more expensive than cellular networks -unsuitable for latency sensitive applications -low data rate	[86]
LEO	-test in TCL3 of the US UTM	A2G	-nm	-nm	[60]
Other networks					
ADS-B	-surveillance -tracking	A2G	-mature technology -acceptable range and accuracy	-does not ensure accepted level of security -not tested in high density of UAVs	[64]
	-test surveillance for the TCL3 of US UTM	A2G	-nm	-nm	[60]
APRS	-transmission of surveillance data	A2G	-provide an altitude more than 4G/LTE	-congestion problem in a high mass of UAVs in conflict	[64]
LoRa	-transmission of surveillance data	A2G	-long-range transmission	-low power -low rate	[64]
	-tactical deconfliction: peer-to-peer scenario	-A2G -U2U	several types of communication: -via ground infrastructure: suitable in pre-flight procedure -peer-to-peer	-loss of messages for distances more than 1km between two UAVs -At 2km, 80% of messages are not received	[97]

Table 9. Autonomy Levels

Autonomy level	Description	Capability	Problems resolution
0	Remotely piloted	the UAV is totally controlled by the operator: direct real time control	loss of communication generates the loss of the UAV+mission failure
1	Remotely guided	the UAV can execute a predefined plan generated by the operator	system failure: the operator recovers the UAV manually
2	Real time health diagnosis	the UAV can do an auto monitoring of its health the UAV can detect by itself system faults	system fault: the operator is responsible for the corrective action
3	Adapt to flight and failure conditions	the UAV can do an auto monitoring of its health the UAV can detect by itself system faults and then re-plan its flight	loss of communication: the UAV restores the link or follows a predefined plan
4	On board route re-plan	does not need to interact with the operator to resolve problems	threat: the UAV re-plans its flight

the mitigation of communication component demands. However, it requires a substantial computational capacity on-board the vehicle. An alternative architecture involves conducting the risk prediction remotely while maintaining the other two functions on-board the vehicle. It allows the resource sharing and a reduced size and weight on-board but the decision-making is dependent on a communication that must be robust. The third architecture is based on remotely risk prediction and path generation with on-board decision-making. In the final architecture all the functions are hosted remotely. Here, the decision-making is widely dependent on the network connection. The selection of one architecture over others is primarily linked to the computational capacity within the vehicle and the resilience of the communication system.

7.2 Centralized vs Decentralized UTM Decision Making

A UTM system bases its functions on a continuous flow of information with the participating entities and especially the operator, the ATM, and the UAV in charge of the operation. In [81], the communication messages are categorized into three classes: (i) Control messages: are sent from the operators to the UAV. They are used to carry the information needed to control and update the UAV trajectory according to the new airspace constraints. (ii) Telemetry messages: include the **Global Navigation Satellite System (GNSS)** position, status, and altitude information. (iii) Awareness messages: contain information about the current position of the UAV and its future trajectory. The UAV autonomy is defined as the ability of a UAV to perform its mission on its own without the operator intervention. It influences directly the communication links needs as well as the basic requirements of the command, telemetry, and awareness messages. In [81], four levels of autonomy are presented that we summarize in Table 9. We also summarize in Table 8, that is provided as online supplementary material, the communication links requirements, knowing that all the values presented in [81], are based on the **International Telecommunication Union (ITU)** report M.2171.

In [22], several models of autonomy levels are presented, but we notice that they revolve around the same idea. The more the level of autonomy increases, the less the UAV will need command and telemetry links with the operator or the UTM. A high level of autonomy means a capacity of self-decision making and the possibility of communication between UAVs to avoid conflicts. For military UAVs, the automation levels must be linked to detection and tracking capabilities and a self tactical deconfliction aptitude.

In a UTM system, there are common information flows, whether the decision-making is centralized or decentralized. The UTM gets from the ATM the current state of the VLL airspace, especially concerning the manned aircraft and the geofences. In turn, the UTM must inform the ATM if there are UAVs which will cross the controlled part of the airspace or will pass around the airports. The operator is also in continuous communication with the UTM to provide necessary information about the UAVs and receive the instructions of the mission status updates and conflict resolution

Table 10. Information Flow in Centralized and Decentralized Decision Making in UTM

Information flow	Communication link	Centralized UTM	Communication message	Distributed UTM	Communication message
Uncommon information flow	UAV to UAV			✓	Awareness
	UAV to UTM	✓	Awareness		
	UAV to operator	✓	Telemetry	✓	Telemetry
Common Information flow	Operator to UAV	✓	Command	✓	Command
	Operator to UTM	✓	Status updates	✓	Status updates
	UTM to operator	✓	-Mission updates -Conflict resolution	✓	Mission updates
	UTM to ATM	✓	Airspace impacts	✓	Airspace Impacts
	ATM to UTM	✓	Airspace usage	✓	Airspace usage

[81]. A UAV sends to the operator, in regular time intervals, telemetry information and then receives orders. However, when the decision-making is centralized, the UTM center is considered as the main entity to resolve conflicts. Each UAV has to transmit continuously awareness messages to the UTM center, leading to a high load on the air-ground communication. The processing of this huge number of awareness data by a single unit remains a real challenge. If a conflict is detected, the information flow from the UTM to the operator contains the updated trajectories, which in turn, transmits the new commands to the UAV. The case is different in a distributed decision-making, the awareness data is no longer transmitted to the UTM, but is exchanged between the UAVs. If a conflict is detected, the concerned UAV makes a self decision or reacts by relying on a consensus mechanism. The direct communication between the UAVs and the local decision-making reduces the decision latency (see Table 10).

In this regard, [99] presents the centralized and decentralized conflict resolution in urban air mobility. The decision-making process is defined almost similarly to [81]. In fact, a ground-based resolution service collects all necessary information from vehicles flying in the sky through surveillance systems and sends new trajectories to vehicles if a conflict is detected. The decision-making is not done in real time, the conflicts are resolved subsequently following their priorities. The conflict resolution service complains about errors that are usually found in the received surveillance data. The tracking and filtering methods (discussed earlier in Section 4.2), decrease the imperfect data, but does not eliminate them completely. In addition to the packet loss issue, which degrades the quality of the air-to-ground communication. The decentralized conflict resolution is a process based on the communication between the UAVs more than the air-to-ground communication. Depending on the degree of collaboration between UAVs, [99] proposes three categories of decentralized decision-making: (i) Decentralized with uniform rules: each UAV resolves conflicts by executing a set of predefined rules and protocols. The communication between the UAVs is only to share necessary data. (ii) Decentralized with coordination: the UAVs coordinate together by a predefined negotiation mechanism in real-time manner. (iii) Decentralized with mixed rules: the UAVs do not follow the same rules in one area, each group of UAVs follow a different set of conflict resolution rules. The decentralized decision-making is not deprived of the imperfect data derived from the **Communications, Navigation, and Surveillance (CNS)** systems, but the on-board conflict resolution and the predefined rules to follow, alleviate this issue.

8 PERFORMANCE EVALUATION

Most of the UTM systems are still under development, a few UTM

and navigation system performance [55], strategic deconfliction performance [52] and USS performance [53]. A **measure of effectiveness (MOE)** is a qualitative evaluation associated to each system requirement. The quantitative aspect is expressed using the **Measures of Performance (MOPs)**. For each MOE are associated one or more MOPs. We give in Table 9, that is provided as online supplementary material, some extracts of the MOPs relating to the strategic deconfliction, USS, and communication and navigation entities. For more details the reader can refer to [52, 53, 55]. The collection of MOPs values is done in the setting of the UTM system in emergency cases, conflicts, and communication loss to study the system reaction in such situations. The FAA collaborated with NASA to perform four **Technical Capability Level (TCL)** demonstrations and two national campaigns. In each TCL, a set of real and simulated operations are achieved with well-defined MOPs measures used to draw lessons necessary to improve the UTM quality of services.

In the same context, [74] defines the safety indicators, which can be used to measure the safety performance of U-space. Determining these indicators is a step following the identification and grouping the risks and hazards. The authors grouped the hazards into categories according to the following factors: (i) The vehicle reliability and its equipment. (ii) The qualification and the performance of the operator. (iii) The environmental factors like the weather. (iv) The airspace usage and the population density. (v) The communication protocols and infrastructure.

The authors in [74] have provided a comprehensive report of the safety indicators for each hazard category where the interested reader can find more details. To test the performance of the blockchain UTM systems, [12] uses cAdvisor. It is a tool essentially employed to evaluate the performance of the blockchain applications. The performance is measured based on resource usage (memory usage, CPU consumption) and the network latency, which is the total time duration taken for the execution of a transaction in the blockchain network. But, it lacks from comparisons with other UTM systems to evaluate the results found. The simulators are useful in the context of the UTM. Actually, it is difficult or even impossible to achieve real testbeds with numerous drones due to cost and time constraints [69]. Testing emergency cases namely the inter UAV collision and collisions between UAVs and the manned aircraft is not possible by real testbeds. Hence, using simulators is more reassuring than putting the UAVs, the manned aircraft or people, in danger. The **Japan Aerospace Exploration Agency (JAXA)** developed the **Scalable Simulator for Knowledge of Low altitude environment (SKALE)**. SKALE is used for the parcel transport use case to evaluate the airspace safety and the efficient airspace utilization [69]. The main features of the simulated UTM are the same as the US UTM: the FIMS, the UASSP, the SDSP, and the ATC. In each simulation, simultaneous flights can be performed where the number of drones can reach 250 with a maximum speed of 15 m/s. The models used for drones are the Multirotor and DJI Matrice 600. For each drone is associated only one GCS. The necessary delay for the FIMS to process the status of the drone and sending instructions is 4 seconds. Whereas, the delay is minimized to 2 seconds if the UASSP processes the status of the drone without considering the FIMS. A **Conflict Detection and Resolution (CDR)** process is applied using the concept of the protected airspace and some measures of separation. The protected airspace is defined in [69] as the distance margin between a flight path and other flight paths. In SKALE, the emergency cases related to weather and malfunction issues are not considered. Millan-Romera et al. present, in [67], a simulation framework based on the **Robot Operating System (ROS)** and the simulator Gazebo. This simulator uses the architecture of U-space. Using ROS, the UTM is modeled as a set of independent nodes. The UTM manager node reflects the aspect of the system, and the DB manager is the node linked to all databases and responsible for the basic write and read operations. The main pre-flight and in-flight services namely the flight plan management, tracking, monitoring, emergency management, and conflict solver of U-space are modeled as nodes too. They are implemented and tested with the ROS-MAGNA test bench. A few number of fixed and rotary wings autonomous UAVs

are used in a real scenario as part of the GAUSS project. However, the scenario is not tested in dense airspace traffic. More complex scenarios of emergency management have to be considered. The decision-making is centralized. Hence, the ground stations are responsible for receiving the instructions, deciding and sending the decisions. UTSim is another simulator for the UTM [13]. It is based on the Unity platform. This platform can support 3D graphics and models the UAVs and their properties as game objects. UTSim took advantage of some Unity components to implement the UAV movements, tracking, and collision detection. Periodic and emergency messages are implemented in different scenarios. The most important one is the scalability and collision scenario in which the number of drones that are allowed to fly simultaneously was varied from 20 to 1,500. The tests are done without using any collision avoidance technique. They concluded that the number of collisions is linearly proportional to the number of drones. Another initiative is the eUTM. The main idea is to be flexible and expandable to support the continuous changes and enhancements of technologies and regulations for the UTM [102]. In this simulator, the main objective is to implement deconfliction algorithms for both VLOS and BVLOS operations. All the simulator development is based on open sources even for the platforms to 3D visualization and supplemental information concerning the terrain and geo-information in real time. The deconfliction is tested with the Barfield's algorithm [20], which is an existing collision avoidance algorithm in the literature, based on geometric method. Some limits are mentioned by the others to be considered as a future work. It is necessary to test the simulator with more deconflicting algorithms and to deal with deconfliction in the integrated airspace between manned and unmanned aircraft [102]. The UTM simulator, defined in [29], is agent-based and has a micro-service architecture. This type of architecture allows separating the behavior of agents in different micro-services. The main elements of this simulator are the drones, the pilots, the GCSs. The surveillance and the ground station network are modeled as an independent agent. The main simulated services are those of the control of the UAV trajectory, the C2 link and the surveillance by using sensors. During the simulation, agents change their states and interact each with the others. One advantage of this simulator is that it considers the positioning errors when using the GNSS systems by adding some measurement error in the real drone position. This position is after filtered with filter techniques.

9 OPEN ISSUES

The process of integration of the UAVs with the manned aircraft in the controlled and uncontrolled airspace raises several challenges. In this section, we identify the issues and challenges that we believe are among the most compelling in the context of the UTM development.

9.1 Interoperability

The interoperability within the UTM system denotes the manner in which data is exchanged and interpreted among the diverse entities participating in the system [45]. In centralized UTMs, the concern of interoperability does not have a significant impact, as there exists a singular central entity responsible for disseminating shared data to all participating entities. However, in decentralized UTMs, where a central unit is absent, data exchange occurs among multiple participants, thus highlighting the crucial challenge of establishing compatible protocols, data format and standards for seamless communication. Besides the architectural considerations, the exchange of information between UTM and ATM is crucial. However, there is a lack of common standardization and protocols to facilitate the exchange of airspace awareness and geofence information among UTM, ATM, and ATC. Therefore, it is necessary to direct research efforts towards developing an interface for the exchange of critical information between UTM and ATM. A UTM can be established within each geographical region, wherein these UTMs engage in communication and exchange of shared data in various scenarios, including when a UAV makes transitions between regions to

execute a mission. In such instances, the lack of compatibility between communication protocols and adherence to common standards among these UTMs poses a significant challenge in attaining interoperability. The current ICAO's classification of the airspace is primarily intended for manned aircraft. Sharing the space with the UAVs requires a reorganization of the ICAO's classification to consider the UAV operations that may be performed in the controlled part.

9.2 Centralized vs Decentralized Architectures

Most of the developed UTM architectures are centralized, with a centralized decision-making as well. The limits of the centralized UTMs are discussed in Section 3.2. The decentralized architectures present a solution for the single point of failure problem. However, this does not prevent the decentralised UTMs from several issues. The decentralized UTM, primarily characterized by the distribution of services among multiple entities, particularly in monitoring and conflict resolution, presents various challenges encompassing safety, interoperability, conflict management, and delays. The sharing of critical data that demands utmost accuracy and reliability among multiple entities introduces safety risks when information is inadequate or the source is not reliable. Moreover, the risk of collisions arises from delays caused by communication loss or disruption between UAVs. Instances of delayed local decision-making among UAVs can have catastrophic consequences. Interoperability also emerges as a significant challenge in decentralized UTMs, necessitating extensive efforts to establish common and compatible protocols and standards. Conflict management within a decentralized UTM heavily relies on telemetry data sharing and local decision-making at the UAV level, requiring advanced DAA technologies and robust communication links. These complexities pose technological challenges at the UAV level. This situation raises inquiries regarding the extent to which a UTM system should incorporate decentralization, employ a hybrid architecture, necessitate a central entity, or adopt a fully distributed structure. The determination of this design choice remains an open issue, serving as a prospective research directive for our future endeavors.

9.3 Massive Data and Security in UTM

UTM must be able to manage and ensure the secure flights of a substantial number of UAVs, thereby necessitating the frequent updating of a considerable volume of data records. However, the limited computing capacity, storage, and processing memory and energy of the UAVs is a problem for both the centralized and decentralized UTM. The UTM must have the capability to manage a great volume of telemetry, command and control data, as well as accommodate different UAV speeds and altitudes. In the same context, the decentralized UTM is essentially based on the communication and information sharing between UAVs. However, due to the aforementioned limitations, lightweight solutions are necessary. Furthermore, the use of distributed databases helps in managing the enormous volume of records. However, it is important to note that sensor and surveillance data often contain imperfections and imprecisions. The UTM relies on this data to make accurate and timely predictions regarding airspace traffic awareness. Therefore, there is a need to enhance filtering methods and surveillance technologies to improve the quality of the data. Managing and securing such a vast volume of UTM data poses significant challenges. With the increasing reliance on data-driven decision-making and communication systems in UTM, there is a higher risk of cyber threats and vulnerabilities. In fact, several cyber-attacks like hijacking, GPS spoofing, and jamming can affect the safe running of the operations and may cause failed missions and even collision risks. The attacks against the GCSs can disrupt the C2 link and the telemetry messages, causing a denial of service. In this case, the hazards can vary from the congestion of the network and unavailability of the UTM to the collision risk. Identity spoofing can compromise UAVs' and GCSs' credentials, enabling the attacker to send falsified messages. Thus, the development of the UTMs must consider the security aspect. It is important to provide secure links and ensure integrity and

privacy protection. Security is a shared concern encompassing both centralized and decentralized UTM. Specifically, the reliance of centralized UTMs upon a singular infrastructure and central entity exposes them to distinctive vulnerabilities. Attacks targeted at a central control center possess the potential to engender system incapacitation. In the context of decentralized systems, risks may hinder the transmission of information among autonomous UAVs, thereby potentially instigating vulnerabilities. The blockchain may be an efficient technology to develop secured UTM systems.

9.4 Communication Links

In the UTM, accuracy, availability, and reliability are of utmost importance to ensure real-time information updates and seamless coordination among the operators, USSs, and the UAVs themselves with their operators. Several communication technologies are being evaluated to select the more efficient ones for fulfilling the communication link requirements of a UTM. Some issues and challenges are identified and have to be addressed. Cellular networks have been designed primarily for terrestrial use. They present some limits due to their limited coverage in 3D environment and availability problem in non-urban areas. Issues related to efficiency, interference and message drops are notably observed, particularly in the context of LTE networks in a high density of UAVs. The cellular networks are required to provide uninterrupted availability to ensure effective collision avoidance which is a challenge especially in rural environments. Dropped links in sudden high demand and C2 link failure are real problems that can lead to collision risks. Therefore, improvements must be implemented to enhance the support of cellular networks for UAV communications. The use of redundant communication systems can help mitigate the risks associated with dropped links. Furthermore, integrating **Non-Terrestrial Networks (NTN)** with terrestrial networks in the upcoming 6G architecture presents a viable solution for establishing a whole connected airspace. Enabling operators to conduct BVLOS operations is a key objective of the UTM and advanced air mobility. These operations encompass activities such as package delivery, traffic management and surveillance, as well as search and rescue missions, all of which necessitate BVLOS communication capabilities. Critical factors for successful BVLOS links include reliability, long-range communication capabilities, and extremely low end-to-end latency. Establishing regulations governing BVLOS communications requires significant efforts and attention to ensure safe and efficient operations in the UTM ecosystem. The A2A links are crucial for the development of decentralized UTMs as well as the distributed decision-making. Specific standards should be developed to support the A2A communication.

9.5 AI Challenges in the Context of UTM

The use of AI in UTM presents open issues related to safety, interpretability, scalability, adaptability and ethical considerations. Effectively addressing these challenges is of utmost importance for the successful integration of AI technologies in UTM and ensuring their effective deployment in managing unmanned aerial traffic [2]. Safety and reliability are of paramount importance in UTM. To operate in controlled airspace UTM systems need to demonstrate high levels of safety and reliability to obtain the necessary approvals, certifications, and authorizations. Using an AI algorithm in a UTM ecosystem needs to meet the high safety standards required by international and national regulators (ICAO, FAA, EASA and **IATA (International Air Transport Association)**). AI models need to be thoroughly tested, validated, and certified to ensure compliance with these standards [59]. In the context of UTM, where transparency and accountability play pivotal roles, the interpretability and explainability of AI algorithms are crucial. However, AI models often operate as opaque systems, making it challenging to comprehend the underlying decision-making process and the rationale behind their outputs. The ability to provide explanations regarding the reasoning and mechanisms behind AI systems' decision-making is essential for international and

national regulators [17, 45]. Scalability and efficiency are open issues in the context of using AI in UTM. AI algorithms must be able to efficiently process and analyze data from multiple sources (UAVs, ATCs, sensors) simultaneously. Developing AI algorithms that can efficiently process large amounts of real-time data from multiple sources, while maintaining low-latency performance, is an ongoing challenge [37]. In UTM, the operating conditions can quickly change in complex and dynamic environments. One of the critical tasks for UTM is collision avoidance, which involves detecting and preventing potential conflicts between UAVs and other objects such as buildings, aircraft, or other UAVs. AI systems must be capable of adapting and learning from changing data to make real-time decisions [48]. Using AI in UTM gives rise to ethical and legal considerations, including issues of accountability for decisions made by AI systems, possible biases, and adherence to privacy regulations. Addressing the ongoing issue of ensuring compliance with legal requirements, ethical principles, and societal concerns within AI systems requires careful consideration and robust frameworks [79].

9.6 Managing Communication Spectrum in the UTM

The spectrum allocated for the UTM comprises a range of radio frequency bands, which are under the control of each respective state. These frequency bands are utilized to facilitate the communication and coordination among all the stakeholders involved in UTM operations. Currently, there is no dedicated spectrum exclusively designated for UTM purposes. The UTM operations share licensed and unlicensed frequencies with other wireless and traditional mobile services. They enable the exchange of crucial data, such as command signals, telemetry information, flight plans, airspace data, and real-time updates. The spectrum for UTM encounters several challenges, including spectrum scarcity and congestion, interference issues, and the ongoing evolution of UTM technologies. The spectrum scarcity: As the utilization of UAVs increases in several applications, the demand for spectrum amplifies, while also considering the imperative of coexisting frequencies with other applications and services. Due to the finite nature of the spectrum resource, the allocation of frequencies for UAV communications is limited, particularly in densely populated and congested regions where the demand for wireless services is high. The dynamic spectrum access can be used to assure fairness and efficient allocation of frequency bands [51]. The interference mitigation: The existing frequency bands are primarily used for traditional mobile services which is for example the case of the LTE frequency range 800-900Mhz used in the same time for the control of UAVs and for GSM systems. This coexistence raises concerns about potential interference that could adversely impact the reliability and performance of UTM communications. Consequently, the development of effective interference mitigation techniques becomes imperative. To address this issue, leveraging AI techniques proves beneficial. AI algorithms can predict regions characterized by high levels of interference based on historical data and real-time measurements. By incorporating these predictions into route planning algorithms, UAVs can identify and plan paths that minimize the impact of interference [77]. Technology evolution: The rapid advancements in the UTM used technologies, particularly in the field of communications, necessitate the fulfillment of new requirements. Specifically, the integration of V2V communication for collision detection purposes, which involves transmitting information such as position and speed, entails the development of suitable standards. This development presents challenges related to the allocation of sufficient spectrum, bandwidth, and security measures to support V2V communication effectively [16].

9.7 Metrics, Real Testbeds and Simulators for UTM

Only a few UTM have been tested and evaluated in real conditions. Even these systems were evaluated using a limited number of UAVs. The development of simulators is essential to implement and test the UTM services. Currently, some efforts are made to create simulators, but they are limited,

especially for emergency management. The comparison between UTM systems is vital to assess system performance, particularly when comparing decentralized UTM systems with centralized ones. The UTM must adhere to a well-defined set of prerequisites encompassing **Quality of Service (QoS)**, which are evaluated using specific metrics. This adherence to QoS standards is paramount for establishing reliable and efficient communication between the UTM and UAVs. Achieving the desired level of performance entails meeting minimum requirements in terms of latency and packet loss. However, challenges arise due to the limited availability of communication infrastructure capable of accommodating the demanding requirements of the UTM. In the context of the UTM, the timely and accurate exchange of information with relevant entities holds significant importance. The freshness of information is a critical aspect and is quantified using the **Age of Information (AoI)** as a performance metric within the communication system [100]. Employing AoI becomes particularly relevant for the UTM, as it allows for assessing the freshness of critical data, such as control and telemetry information exchanged between the GCS and UAVs. Moreover, fairness represents an ongoing challenge for the UTM. As UAV traffic increases, the UTM aims to alleviate congestion by implementing ground delays. While this improves efficiency, it concurrently decreases fairness. Several parameters contribute to the challenge of achieving fairness, including dynamic traffic demand and the preferences of operators [30]. In addition to addressing congestion, a UTM must consider a crucial parameter known as its capacity. The capacity of a UTM refers to the maximum number of UAVs that can operate simultaneously within a specific geographical area during a defined time interval. Determining the capacity involves considering various factors, including regulatory constraints, frequency availability, and the size of safety zones surrounding the UAVs [84]. Accurately estimating capacity poses challenges, particularly in urban environments, where multiple factors influence the calculation. The estimation of capacity must satisfy two fundamental QoS requirements: safety and performance. Enhancing safety measures may result in restricted flight operations and an extension of travel distance for UAVs, subsequently impacting the overall performance [84]. Striking the right balance between these two requirements poses a significant challenge, as optimizing safety while maintaining acceptable performance levels requires careful consideration and trade-offs.

10 CONCLUSION

The UTM is an active area of research. Its importance is proliferated from the need to develop a system able to manage and ensure secure flights of millions of UAVs sharing the sky with the manned aircraft. In this paper, we presented a comprehensive review of almost all the relevant topics related to the UTM. We focused on the UTM architectures presented in the literature. We proposed to classify the architectures into a novel taxonomy as centralized and decentralized with an exhaustive comparison. Also, we classified the centralized UTM architectures into sub-categories: based on regular rules, hierarchical, service oriented and based on a specific cellular network. A UTM must offer a set of services and we detailed the most important ones. We studied in depth the registration and the identification services, we presented the used technologies and discussed their limits. We detailed the main features of the tracking, surveillance and monitoring services. We subsequently detailed the collision avoidance service which is the more critical and is mainly related to the information delivered from the tracking service and alerts from the monitoring service. We presented the process of the conflict management, the different approaches and technologies used for detecting and resolving conflicts. At the same time, we detailed the communication requirements necessary to satisfy the UTM communication links including latency, reliability and availability. The level of UAV autonomy directly impacts the decision-making process of the UTM. Therefore, we have presented both centralized and distributed decision-making approaches. The decision-making process in emergency cases is of paramount importance and should be given the

highest priority. Thus, we listed the taxonomy related to the emergency and discussed the different approaches used to deal with this issue. The UTM systems are still in an early stage of development, with only a minority being tested under real conditions and none of them being compared with other UTMs using quantitative metrics. There are no clear measures to assess the performance of UTMs. We presented in this survey, the MOPs used by NASA to evaluate the UTM. We listed the simulators presented in the literature used to test the UTMs. Lastly, we have identified various issues and highlighted the areas that demand further research and exploration. These include the spectrum allocation for the UTM, communication links, architectural design choices, utilization of AI algorithms in the UTM, management of large-scale data in the UTM, interoperability, and the comparison of UTMs using quantitative metrics. This survey provides a very detailed study of all the concepts related to UTMs and can help researchers to have a clear view of all the issues involved.

SUPPLEMENTARY ONLINE MATERIAL:

Supplementary online material contains tables depicting summaries and explanations for some concepts presented within this document.

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